A well-balanced and positivity-preserving numerical scheme for the Ripa system with wet-dry fronts

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Abstract

In this paper, we propose a second-order well-balanced finite-volume scheme for the one- and two-dimensional Ripa system in the presence of wet-dry fronts. The method relies on a new pre-balanced reformulation in which the pressure-gradient term is decomposed into a flux contribution and a source term, allowing an exact discrete preservation of the lake-at-rest equilibrium with wet-dry areas and the isobaric steady state. To eliminate spurious pressure oscillations across temperature discontinuities in isobaric equilibria, the scheme combines a balanced source discretization with a positivity-preserving modification of the Harten–Lax–van Leer (HLL) flux adapted from the literature, in which the intermediate-wave contribution is scaled by a suitable parameter. High-order accuracy is achieved through a MUSCL reconstruction with a generalized minmod limiter, applied either to primitive or equilibrium variables depending on the targeted steady state, together with hydrostatic reconstruction to ensure non-negativity of the water depth and temperature. Time integration is performed using the second-order strong stability preserving Runge-Kutta method (SSP-RK2) under a suitable CFL condition. Numerical experiments in one and two space dimensions confirm second-order accuracy, preservation of the steady states up to machine precision, and robust, oscillation-free behavior near wet-dry interfaces and temperature discontinuities.

Keywords: Ripa system; Well-balanced scheme; Wet-dry fronts; Finite-volume scheme; Steady states.

1 Introduction

We consider the modification of the shallow water equations taking into account water temperature fluctuations. In one space dimension, the system reads as follows

$$\begin{cases} \partial_t h + \partial_x(hu) = 0 \ , \\ \\ \partial_t(hu) + \partial_x\Big(hu^2 + \dfrac{gh^2\theta}{2}\Big) = -gh\theta\partial_x b \ , \\ \\ \partial_t(h\theta) + \partial_x(hu\theta) = 0, \end{cases} \tag{1.1}$$

where $h(t,x)$ denotes the water depth, $u(x,t)$ the depth-averaged velocity, θ the potential temperature field, g the gravitational acceleration, and $b(x)$ the bottom topography, which is assumed to be time-independent. This system was introduced in [1, 2, 3] for modeling ocean currents and is commonly referred to as the Ripa system. It can be obtained by depth-averaging multilayer ocean models.

The steady-state solutions of (1.1) satisfy

$$\begin{cases} \partial_x(hu) = 0 \ , \\ \partial_x\Big(hu^2 + \dfrac{gh^2\theta}{2}\Big) = -gh\theta\partial_x b \ , \\ \partial_x(hu\theta) = 0. \end{cases} \tag{1.2}$$

In general, system (1.2) does not admit a closed-form characterization of all steady-state solutions. Nevertheless, several important steady states can be identified. Among them are the following three still-water equilibria.

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The constant free-surface (lake-at-rest) steady state is given by

$$\theta \equiv \text{constant}, \quad h + b \equiv \text{constant}, \quad u \equiv 0, \tag{1.3}$$

which follows directly from (1.2).

The isobaric steady state reads

$$b \equiv \text{constant}, \quad \frac{g}{2}h^2\theta \equiv \text{constant}, \quad u \equiv 0, \tag{1.4}$$

and is obtained from (1.2).

Finally, the constant water height steady state is characterized by

$$h \equiv \text{constant}, \quad b + \frac{h}{2}\ln\theta \equiv \text{constant}, \quad u \equiv 0. \tag{1.5}$$

The steady states (1.2) also include a smooth moving steady state, given by

$$hu \equiv \text{constant}, \ \frac{u^2}{2} + g\theta(h+b) \equiv \text{constant}, \ \theta \equiv \text{constant}. \tag{1.6}$$

In this work, we mainly focus on preserving the steady states (1.3)-(1.4). Many physically relevant flows, including lake surface waves and deep-ocean tsunamis, may be characterized as small perturbations of these steady states. Accurately resolving such dynamics is challenging unless the numerical scheme preserves the underlying unperturbed equilibrium at the discrete level. The Ripa system exhibits a delicate coupling between the hydrostatic pressure and the advected temperature field. In particular, the pressure term depends nonlinearly on both the water depth and the temperature, and the source term involves their interaction with the bottom topography. This structure makes the preservation of hydrostatic equilibria significantly more intricate than for the classical shallow-water equations. The presence of temperature discontinuities and wet-dry fronts further complicates the design of stable and accurate numerical schemes.

Over the past decade, substantial effort has been devoted to the development of well-balanced numerical schemes for the Ripa system capable of preserving the steady states (1.3)-(1.6) at the discrete level. Early contributions include the work of Chertock et al. [4], who proposed a well-balanced and positivity-preserving central-upwind scheme for the one- and two-dimensional Ripa system, incorporating an interface-tracking technique to suppress spurious pressure oscillations near temperature discontinuities while exactly preserving lake-at-rest steady states. Unstaggered central schemes with well-balanced properties were proposed in [5, 6]. Desveaux et al. [7] introduced a relaxation-based formulation capable of preserving a broad class of hydrostatic equilibria. High-order accuracy was achieved by Han and Li [8], who designed well-balanced finite-difference WENO schemes and extended their methodology to two dimensions. A well-balanced, positivity-preserving, and entropy-dissipative HLLC scheme for the Ripa system was derived in [9], within the framework of path-conservative approximate Riemann solvers. Kinetic-based constructions were investigated in [10] for both one- and two-dimensional models, while a space-time CESE approach was subsequently developed in [11], yielding a second-order accurate, positivity-preserving, and well-balanced scheme. Riemann problems for the Ripa system with discontinuous bottom topography were investigated in [12]. Britton and Xing [13] presented a well-balanced discontinuous Galerkin (DG) method for the Ripa model to maintain the moving-water equilibria, separating the solution into the equilibrium and fluctuation components with appropriate approximations of the numerical flux and source terms. More recently, a well-balanced and positivity-preserving surface reconstruction scheme has been developed in [14] maintaining smooth equilibria, including the still-water and moving-water steady-state solutions.

All the aforementioned literature on Ripa systems does not address the treatment of wet-dry fronts which may occur within the interior of a cell. However, an appropriate treatment of wet-dry fronts is essential for accurately capturing wetting and drying processes. The number of works addressing wet-dry fronts for the Ripa system remains very limited compared with the extensive literature on well-balanced wet-dry treatments for the classical shallow-water equations, see, e.g. [15, 16, 17, 18, 19, 20, 21] for a few references. Recently, Wang and Chen [22] developed a positivity-preserving and well-balanced wet-dry front reconstruction for the Ripa system. Their approach combines a conservative-variable reconstruction, which increases the CFL number in fully flooded cells, with a draining-time correction of the upwind flux in partially flooded cells. Their scheme preserves the lake-at-rest steady state, while preservation of the isobaric steady state is not addressed. In this work, we present a second-order finite-volume scheme for the

Ripa system with wet-dry fronts that exactly preserves lake-at-rest configurations with dry areas and the nontrivial isobaric equilibrium, while remaining positivity-preserving and pressure-oscillation-free across temperature jumps. The achievement of these properties relies on two main ingredients. The first is a new pre-balanced reformulation in terms of the free-surface elevation above datum, which reveals a flux-source structure naturally adapted to the temperature-dependent character of the model. This formulation allows the equilibrium structure to be embedded directly into the governing equations, so that in wet-bed configurations no special source-term treatment is required, while in wetting and drying situations, the balance is recovered through a consistent reconstruction of the Riemann states together with additional balancing terms, without modifying numerical fluxes or clipping flow variables during the simulation. The second ingredient is an appropriate treatment of the temperature-dependent contribution arising from the nonlinear coupling between the water depth and the potential temperature in the Ripa model. This term alters the classical hydrostatic balance structure and must be handled carefully to preserve the isobaric steady state (1.4) at the discrete level. To this end, a consistent source-term discretization, combined with the positivity-preserving Harten, Lax, and van Leer (HLL) flux modification of [14], ensures preservation of the isobaric steady state and prevents spurious pressure oscillations across temperature discontinuities.

The rest of the paper is organized as follows. In Section 2, we introduce the new pre-balanced reformulation of the Ripa system and discuss its structural properties. Section 3.1 presents the finite-volume discretization in one space dimension, including the construction of the numerical fluxes, the source-term discretization, and the positivity-preserving modification of the HLL flux. The well-balanced properties for the lake-at-rest and isobaric steady states are established in this section. In Section 3.2, we extend the scheme to two spatial dimensions and state the corresponding CFL condition. Section 4 is devoted to numerical experiments in one and two dimensions. Finally, Section 5 summarizes the main conclusions and outlines directions for future research.

2 Pre-balanced formulation

In what follows, we derive a new formulation of the Ripa system (1.1) in which the surface gradient term is expressed in terms of the water level above a fixed datum. For this purpose, we introduce the variable $\eta = h + b$, denoting the interface elevation relative to this datum. Using the substitution $h = \eta - b$, it is important to highlight that

$$\partial_x\Big(\frac{gh^2\theta}{2}\Big) + gh\theta\partial_x b = \partial_x\Big(\frac{g}{2}(\eta^2 - 2\eta b)\theta\Big) + g\eta\theta\partial_x b + \frac{g}{2}b^2\partial_x\theta. \tag{2.1}$$

Now, taking into account the time-independence of the bottom topography, the Ripa system (1.1) can be rewritten in an equivalent form. In this formulation, the first equation governs the evolution of the free-surface elevation η, while the second equation is obtained using the identity (2.1). The resulting reformulated Ripa system reads

$$\begin{cases} \partial_t\eta + \partial_x(hu) = 0\ , \\ \partial_t(hu) + \partial_x(hu^2) + \partial_x\Big(\dfrac{g}{2}(\eta^2 - 2\eta b)\theta\Big) = -g\,\eta\theta\,\partial_x b - \dfrac{g}{2}b^2\partial_x\theta\ , \\ \partial_t(h\theta) + \partial_x(hu\theta) = 0\ . \end{cases} \tag{2.2}$$

In matrix form, the Ripa system (2.2) can be expressed as:

$$\partial_t\mathbf{U} + \partial_x\mathbf{F}(\mathbf{U}, b) = \mathbf{S}(\mathbf{U}, b), \tag{2.3}$$

with

$$\mathbf{U} = \begin{pmatrix}\eta\\ q\\ p\end{pmatrix} = \begin{pmatrix}\eta\\ hu\\ h\theta\end{pmatrix},\quad \mathbf{F}(\mathbf{U}, b) = \begin{pmatrix} q \\ \dfrac{q^2}{h} + \dfrac{g}{2}(\eta^2 - 2\eta b)\dfrac{p}{h} \\ \dfrac{qp}{h}\end{pmatrix},\quad \mathbf{S}(\mathbf{U}, b) = \begin{pmatrix}0\\ S_1 + S_2\\ 0\end{pmatrix}, \tag{2.4}$$

where

$$S_1 := -g\eta\frac{p}{h}\partial_x b \quad\text{and}\quad S_2 := -\frac{g}{2}b^2\partial_x\Big(\frac{p}{h}\Big). \tag{2.5}$$

The Jacobian matrix is defined by

$$\mathbf{J}(\mathbf{U},b) = \frac{\partial \mathbf{F}(\mathbf{U},b)}{\partial \mathbf{U}} = \begin{pmatrix} 0 & 1 & 0 \\ -\dfrac{q^2}{h^2} + \dfrac{g}{2}p\Big(2 - \dfrac{\eta^2 - 2\eta b}{h^2}\Big) & \dfrac{2q}{h} & \dfrac{g(\eta^2 - 2\eta b)}{2h} \\ -\dfrac{qp}{h^2} & \dfrac{p}{h} & \dfrac{q}{h} \end{pmatrix}. \tag{2.6}$$

The system is hyperbolic, provided that $h\theta > 0$ and the eigenvalues of the Jacobian matrix (2.6) are given by $u \pm \sqrt{gh\theta}$ and u. To the best of our knowledge, such a pre-balanced reformulation has not been explicitly derived for the Ripa system. By employing the water surface elevation above datum as a primary flow variable, the resulting formulation preserves the hyperbolic structure of the original system while revealing a flux–source decomposition that is balanced at the continuous level. At the discrete level, the lake-at-rest steady state (1.3) is maintained in the wet-bed case when a consistent discretization of $\partial_x b$ is employed in (2.3). This preservation property is extended here to dry-bed configurations following the approach of [18], adapted to the Ripa system. A key structural difference from the pre-balanced shallow water formulation of [17] is the additional temperature-dependent contribution $-\frac{g}{2}b^2\partial_x\theta$, arising from the nonlinear coupling between water depth and potential temperature in the Ripa model. This term modifies the classical hydrostatic balance structure and necessitates an appropriate treatment for the discrete preservation of the isobaric steady state (1.4). This is achieved through a proper discretization of the temperature-dependent source component combined with the positivity-preserving parameter introduced in [14]. This approach prevents spurious pressure oscillations while maintaining non-negativity of the water depth and temperature.

3 Numerical methods

3.1 One-dimensional scheme

We begin by discretizing system (2.2). Here, L represents the spatial extent of the numerical domain. Let $N \in \mathbb{N}^*$, and for each $i \in [1, N]$, the cells are represented by $m_i = (x_{i-1/2}, x_{i+1/2})$. We define $x_i = \frac{x_{i-1/2}+x_{i+1/2}}{2}$ and $\Delta x = x_{i+1/2} - x_{i-1/2}$ as the spatial grid. The grid interfaces at $x_{1/2}$ and $x_{N+1/2}$ correspond to the left and right ends, respectively (see Figure 1). Additionally, we consider a time discretization t^n, where $t^{n+1} = t^n + \Delta t$ and Δt represents the time step. Applying a Godunov-

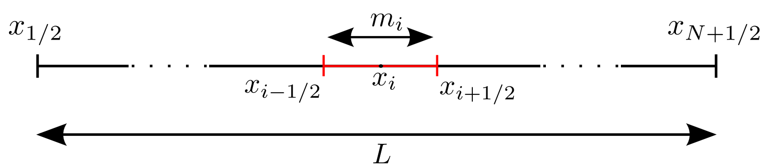


Figure 1: One-dimensional discretization of the computational domain. Each control volume m_i is bounded by the interfaces $x_{i-\frac{1}{2}}$ and $x_{i+\frac{1}{2}}$, and x_i denotes the cell center.

type scheme within a finite volume method, based on equation (2.3), we update in the following the conservative flow variables at each time step:

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x}(\mathbf{F}_{i+1/2}^n - \mathbf{F}_{i-1/2}^n) + \Delta t \mathbf{S}_i, \tag{3.1}$$

where $\mathbf{F}_{i+1/2}$ and $\mathbf{F}_{i-1/2}$ are the fluxes through the east and west interfaces of cell i. The evaluated source terms at the cell centers are denoted by $\mathbf{S}_i$. We use $\mathbf{U}_i^n = (\eta_i^n, q_i^n, p_i^n)$ to represent approximation of $\mathbf{U}$ at the center of cell m_i at time t^n:

$$\mathbf{U}_i^n \approx \frac{1}{\Delta x}\int_{m_i} \mathbf{U}(t^n, x)\, dx\ .$$

When employing piecewise constant approximations for the numerical flux, equation (3.1) represents a first-order Godunov-type scheme, which has the disadvantage of being highly diffusive.

In this study, we expand upon equation (3.1) by developing a second-order scheme in both spatial and temporal dimensions.

3.1.1 Second-order scheme

Employing a monotonic upstream scheme for conservation laws (MUSCL) [23], we perform a linear reconstruction of the variables $\mathbf{V} = \left(\eta, q = hu, \theta = \dfrac{p}{h}\right)^\top$ rather than the conservative variables $\mathbf{U}$. In particular, reconstructing the temperature θ directly with a generalized minmod limiter ensures that the reconstructed face values satisfy a local maximum principle. As a consequence, provided that the cell averages θ_i^n are nonnegative, the reconstructed temperature at cell interfaces remains nonnegative. Moreover, this choice of reconstructed variables is consistent with the preservation of the lake-at-rest steady-state solution (1.3). The reconstructed face values are then used to compute the numerical fluxes.

To avoid oscillations, we adopt the generalized minmod limiter [23, 24] in the construction of the slope limiting function $\partial_x \mathbf{V}_i^n$:

$$\partial_x \mathbf{V}_i^n = \text{minmod}\left(\gamma\frac{\mathbf{V}_i^n - \mathbf{V}_{i-1}^n}{\Delta x}, \frac{\mathbf{V}_{i+1}^n - \mathbf{V}_{i-1}^n}{2\Delta x}, \gamma\frac{\mathbf{V}_{i+1}^n - \mathbf{V}_i^n}{\Delta x},\right), \quad \gamma \in [1,2], \tag{3.2}$$

where

$$\text{minmod}(a,b,c) = \begin{cases} \max(a,b,c) & \text{if } \max(a,b,c) < 0, \\ \min(a,b,c) & \text{if } \min(a,b,c) > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.3}$$

and γ is a minmod limiter that controls the numerical viscosity of the numerical scheme.

On the cell $m_i = (x_{i-1/2}, x_{i+1/2})$, the solution is approached by:

$$\mathbf{V}^n(x) = \mathbf{V}_i^n + (x - x_i)\partial_x \mathbf{V}_i^n.$$

The reconstructed face values of the variables $\mathbf{V}$ are given by (e.g. at the cell interface $i + 1/2$):

$$\begin{cases} \mathbf{V}_i^{n,+} &= \mathbf{V}_i^n + \dfrac{\Delta x}{2}\partial_x \mathbf{V}_i^n \\ \mathbf{V}_{i+1}^{n,-} &= \mathbf{V}_{i+1}^n - \dfrac{\Delta x}{2}\partial_x \mathbf{V}_{i+1}^n. \end{cases} \tag{3.4}$$

The face values of the water depth must be also rebuilt using the same approach:

$$\begin{cases} h_i^{n,+} &= h_i^n + \dfrac{\Delta x}{2}\partial_x h_i^n, \\ h_{i+1}^{n,-} &= h_{i+1}^n - \dfrac{\Delta x}{2}\partial_x h_{i+1}^n, \end{cases} \tag{3.5}$$

Consequently, the bottom topography face values are constructed by subtracting the water depth from the interface elevation above datum:

$$\begin{cases} b_i^{n,+} &= \eta_i^{n,+} - h_i^{n,+}, \\ b_{i+1}^{n,-} &= \eta_{i+1}^{n,-} - h_{i+1}^{n,-}. \end{cases} \tag{3.6}$$

The face values of the velocity are given by:

$$\begin{cases} u_i^{n,+} &= \dfrac{q_i^{n,+}}{h_i^{n,+}}, \\ u_{i+1}^{n,-} &= \dfrac{q_{i+1}^{n,-}}{h_{i+1}^{n,-}}. \end{cases} \tag{3.7}$$

Whenever h vanishes in a dry cell ($h < H_{dry} = 10^{-9}$), the velocities cannot be computed by (3.7). To this end, in dry cells, the face values of the velocities are set to zero. As in [18], the above reconstruction is applied everywhere except at wet-dry interfaces and dry areas. In such case, the face values of the variables $\mathbf{V} = (\eta, q, \theta)^\top$ and bed elevation are assumed to be similar to their corresponding values at the center. Around wet-dry fronts, the second-order accurate scheme is reduced to first-order. This is normal in a slope limiting reconstruction.

To ensure the non-negativity of the water depth, we use a reconstruction technique suggested by Audusse et al. in [16]. Accordingly, the bottom topography at each cell interface (e.g. at $i+1/2$) is defined as follows:

$$b_{i+1/2}^{n} = \max(b_i^{n,+}, b_{i+1}^{n,-}) \tag{3.8}$$

To maintain the positivity of the Riemann states for the water depth at both cell interface sides, we utilize a hydrostatic reconstruction (HR) approach:

$$h_i^{*,+} = \max(0, \eta_i^{n,+} - b_{i+1/2}^{n}), \quad h_{i+1}^{*,-} = \max(0, \eta_{i+1}^{n,-} - b_{i+1/2}^{n}). \tag{3.9}$$

We note that the classical hydrostatic reconstruction has been extended to a subcell hydrostatic reconstruction in [25], which provides an improved treatment of discontinuous bottom topographies. Adapting this construction to the present temperature-dependent system would require subcell discretization of the source terms while retaining positivity and the exact preservation of both the lake-at-rest and isobaric steady states, and is therefore left for future investigation.

The reconstruction of the Riemann states for the conservative variables η, q and p can be expressed in the following way:

$$\begin{cases} \eta_i^{*,+} &= h_i^{*,+} + b_{i+1/2}^{n}, \quad \eta_{i+1}^{*,-} &= h_{i+1}^{*,-} + b_{i+1/2}^{n}, \\ q_i^{*,+} &= h_i^{*,+} u_i^{n,+}, \quad q_{i+1}^{*,-} &= h_{i+1}^{*,-} u_{i+1}^{n,-}, \\ p_i^{*,+} &= h_i^{*,+} \theta_i^{n,+}, \quad p_{i+1}^{*,-} &= h_{i+1}^{*,-} \theta_{i+1}^{n,-}. \end{cases} \tag{3.10}$$

Using a similar process, one obtains the cell interface $i-1/2$ face values.

3.1.2 The numerical flux

The Harten, Lax, and van Leer approximate Riemann solver, described in [26], is adopted to calculate the interface fluxes $\mathbf{F}_{i+1/2}^{n}$ and $\mathbf{F}_{i-1/2}^{n}$ using the aforementioned Riemann states:

$$\mathbf{F}_{i+1/2}^{n} = \widetilde{\mathbf{F}}(\mathbf{U}_i^{*,+}, \mathbf{U}_{i+1}^{*,-}, b_{i+1/2}^{n}) \quad \text{and} \quad \mathbf{F}_{i-1/2}^{n} = \widetilde{\mathbf{F}}(\mathbf{U}_{i-1}^{*,+}, \mathbf{U}_{i}^{*,-}, b_{i-1/2}^{n}).$$

The right interface flux reads as follows:

$$\mathbf{F}_{i+1/2}^{n} = \begin{cases} \mathbf{F}(\mathbf{U}_i^{*,+}, b_{i+1/2}^{n}) & \text{if } a_i^{+} \geq 0, \\ \dfrac{a_{i+1}^{-}\mathbf{F}(\mathbf{U}_i^{*,+}, b_{i+1/2}^{n}) - a_i^{+}\mathbf{F}(\mathbf{U}_{i+1}^{*,-}, b_{i+1/2}^{n}) + a_{i+1}^{-}a_i^{+}(\mathbf{U}_{i+1}^{*,-} - \mathbf{U}_i^{*,+})}{a_{i+1}^{-} - a_i^{+}} & \text{if } a_i^{+} \leq 0 \leq a_{i+1}^{-}, \\ \mathbf{F}(\mathbf{U}_{i+1}^{*,-}, b_{i+1/2}^{n}) & \text{if } a_{i+1}^{-} \leq 0, \end{cases} \tag{3.11}$$

where a_{i+1}^{-} and a_i^{+} are the nonnegative and nonpositive estimates of the left and right wave-speeds respectively obtained from the largest and the smallest eigenvalues of the Jacobian $\mathbf{J}(\mathbf{U}, b)$ in (2.6):

$$\begin{aligned} a_{i+1}^{-} &= \max\left\{u_i^{+} + \sqrt{g h_i^{*,+}\theta_i^{+}},\, u_{i+1}^{-} + \sqrt{g h_{i+1}^{*,-}\theta_{i+1}^{-}},\, 0\right\}, \\ a_i^{+} &= \min\left\{u_i^{+} - \sqrt{g h_i^{*,+}\theta_i^{+}},\, u_{i+1}^{-} - \sqrt{g h_{i+1}^{*,-}\theta_{i+1}^{-}},\, 0\right\}. \end{aligned} \tag{3.12}$$

The calculation of the left interface flux $\mathbf{F}_{i-1/2}^{n}$ is done using similar formulas.

Applying (3.1), we first compute an intermediate solution $\widetilde{\boldsymbol{U}}_i$ using a forward Euler step:

$$\widetilde{\mathbf{U}}_i = \mathbf{U}_i^{n} - \Delta t\, \mathbf{K}_i(\mathbf{U}^{n}), \tag{3.13}$$

where the discrete operator $\mathbf{K}_i$ is defined by

$$\mathbf{K}_i = \frac{\mathbf{F}_{i+1/2}^{n} - \mathbf{F}_{i-1/2}^{n}}{\Delta x} - \mathbf{S}_i. \tag{3.14}$$

The second-order discretization in space is coupled with a second-order *strong stability preserving* Runge-Kutta (SSP-RK2) discretization in time [27]. Accordingly, the fully discrete MUSCL-SSP-RK2 scheme is given by

$$\mathbf{U}_i^{n+1} = \frac{1}{2}\mathbf{U}_i^n + \frac{1}{2}\Big(\widetilde{\mathbf{U}}_i - \Delta t\, \mathbf{K}_i(\widetilde{\mathbf{U}})\Big). \tag{3.15}$$

In this formulation, the update to the new time level is obtained as a convex combination of forward Euler steps.

Remark 3.1. Even though h_i^n is non-negative, it might be very small or even zero. This may be troublesome in computing the temperature from the conserved variables, since we need to divide by small values. To avoid the division by small numbers, we follow [4, 28] and desingularize the division by replacing it with

$$\theta_i = \min\{\theta_{\max}, \max\{\theta_{\min}, R_i\}\},$$

where $R_i = \dfrac{\sqrt{2}h_i p_i}{\sqrt{h_i^4 + \max\{h_i^4, \varepsilon^4\}}}$, and $\theta_{\max} = \max\limits_{i=1,N} \theta_i^n$, $\theta_{\min} = \min\limits_{i=1,N} \theta_i^n$. Here, ε is a small positive number chosen to be Δx in all of our numerical experiments. This choice makes the desingularization threshold proportional to the spatial resolution, so that the regularization is activated only when the water depth becomes comparable to, or smaller than, the mesh size, while ensuring that the threshold tends to zero under mesh refinement.

3.1.3 Positivity preserving property

In what follows, we prove the positivity-preserving property of the proposed numerical scheme.

Theorem 3.2. *(positivity-preserving property) Assume that the system* (3.1) *is discretized using a first-order forward Euler step in time. Assuming $h_i^n \geq 0$, $\theta_i^n \geq 0$ for all $i \in [1, N]$, and the CFL condition satisfies*

$$\Delta t \leq \frac{\Delta x}{4a}, \quad \textit{where } a = \max\big(\max_i(a_{i+1}^-, -a_i^+)\big), \tag{3.16}$$

with a_{i+1}^- and a_i^+ defined in (3.12)*, then we obtain $h_i^{n+1} \geq 0$, $\theta_i^{n+1} \geq 0$ for all $i \in [1, N]$ using the numerical scheme* (3.1)*,* (3.11) *and* (3.21)*.*

Proof. We start first by proving that the water height $h_i^{n+1} \geq 0$. Using (3.9), we have

$$0 \leq h_i^{*,+} \leq h_i^{n,+}, \quad 0 \leq h_{i+1}^{*,-} \leq h_{i+1}^{n,-}.$$

The cell centered value of the water depth satisfies

$$h_i^n = \frac{h_i^{n,-} + h_i^{n,+}}{2} \geq \frac{h_i^{*,-} + h_i^{*,+}}{2}. \tag{3.17}$$

Using the numerical flux (3.11) and the reconstructed water surface elevation in (3.10), the first component of the numerical flux at the cell interface $i + \frac{1}{2}$ is as follows

$$\begin{aligned}
\mathbf{F}_{i+\frac{1}{2}}^{(1)} &= \frac{a_{i+1}^-\, h_i^{*,+}\, u_i^{n,+}}{a_{i+1}^- - a_i^+} - \frac{a_i^+\, h_{i+1}^{*,-}\, u_{i+1}^{n,-}}{a_{i+1}^- - a_i^+} + \frac{a_{i+1}^- a_i^+}{a_{i+1}^- - a_i^+}(\eta_{i+1}^{*,-} - \eta_i^{*,+}) \\
&= h_i^{*,+} a_{i+1}^- \left(\frac{u_i^{n,+} - a_i^+}{a_{i+1}^- - a_i^+}\right) - h_{i+1}^{*,-} a_i^+ \left(\frac{u_{i+1}^{n,-} - a_{i+1}^-}{a_{i+1}^- - a_i^+}\right).
\end{aligned} \tag{3.18}$$

According to the first component of (3.1), together with the forward Euler temporal discretization , it follows that

$$\eta_i^{n+1} = \eta_i^n - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_{i+\frac{1}{2}}^{(1)} - \mathbf{F}_{i-\frac{1}{2}}^{(1)}\right) = h_i^n + b_i - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_{i+\frac{1}{2}}^{(1)} - \mathbf{F}_{i-\frac{1}{2}}^{(1)}\right) = \frac{h_i^{n,-} + h_i^{n,+}}{2} + b_i - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_{i+\frac{1}{2}}^{(1)} - \mathbf{F}_{i-\frac{1}{2}}^{(1)}\right).$$

Then, according to (3.17) and using the fact that $h_i^{n+1} = \eta_i^{n+1} - b_i$, it follows that

$$\begin{aligned}
h_i^{n+1} \geq \frac{h_i^{*,-} + h_i^{*,+}}{2} - \frac{\Delta t}{\Delta x}\Bigg[& h_i^{*,+} a_{i+1}^- \left(\frac{u_i^{n,+} - a_i^+}{a_{i+1}^- - a_i^+}\right) - h_{i+1}^{*,-} a_i^+ \left(\frac{u_{i+1}^{n,-} - a_{i+1}^-}{a_{i+1}^- - a_i^+}\right) \\
& - h_{i-1}^{*,+} a_i^- \left(\frac{u_{i-1}^{n,+} - a_{i-1}^+}{a_i^- - a_{i-1}^+}\right) + h_i^{*,-} a_{i-1}^+ \left(\frac{u_i^{n,-} - a_i^-}{a_i^- - a_{i-1}^+}\right)\Bigg].
\end{aligned}$$

Rearranging the terms yields

$$\begin{aligned}
h_i^{n+1} \quad &\geq \quad h_i^{*,+}\left[\frac{1}{2}-\frac{\Delta t}{\Delta x}a_{i+1}^{-}\left(\frac{u_i^{n,+}-a_i^{+}}{a_{i+1}^{-}-a_i^{+}}\right)\right] \\
&\quad + h_i^{*,-}\left[\frac{1}{2}+\frac{\Delta t}{\Delta x}a_{i-1}^{+}\left(\frac{a_i^{-}-u_i^{n,-}}{a_i^{-}-a_{i-1}^{+}}\right)\right] \\
&\quad - \frac{\Delta t}{\Delta x}h_{i+1}^{*,-}a_i^{+}\left(\frac{a_{i+1}^{-}-u_{i+1}^{n,-}}{a_{i+1}^{-}-a_i^{+}}\right) \\
&\quad + \frac{\Delta t}{\Delta x}h_{i-1}^{*,+}a_i^{-}\left(\frac{u_{i-1}^{n,+}-a_{i-1}^{+}}{a_i^{-}-a_{i-1}^{+}}\right).
\end{aligned} \tag{3.19}$$

From the wave-speed estimates (3.12), we have $a_{i+1}^{-}\geq 0$, $a_i^{+}\leq 0$, and $u_i^{n,+}-a_i^{+}\geq 0$, $a_{i+1}^{-}-u_{i+1}^{n,-}\geq 0$ (and similarly $u_{i-1}^{n,+}-a_{i-1}^{+}\geq 0$, $a_i^{-}-u_i^{n,-}\geq 0$). Moreover,

$$0\leq \frac{u_i^{n,+}-a_i^{+}}{a_{i+1}^{-}-a_i^{+}}\leq 1, \qquad 0\leq \frac{a_{i+1}^{-}-u_{i+1}^{n,-}}{a_{i+1}^{-}-a_i^{+}}\leq 1,$$

so that, under the CFL condition (3.16),

$$\frac{\Delta t}{\Delta x}\,a_{i+1}^{-}\frac{u_i^{n,+}-a_i^{+}}{a_{i+1}^{-}-a_i^{+}}\leq \frac{\Delta t}{\Delta x}\,a\leq \frac{1}{4}\leq\frac{1}{2},$$

and the analogous bound holds for the left-interface coefficient. Therefore, following [4, 14], all coefficients in (3.19) are nonnegative, and hence $h_i^{n+1}\geq 0$.

It remains to prove that the temperature θ_i^{n+1} is nonnegative. Using the numerical flux (3.11), the third component of the numerical flux at the cell interface $i+\frac{1}{2}$ is as follows

$$\mathbf{F}^{(3)}_{i+\frac{1}{2}} = h_i^{*,+}\theta_i^{n,+}a_{i+1}^{-}\left(\frac{u_i^{n,+}-a_i^{+}}{a_{i+1}^{-}-a_i^{+}}\right)-h_{i+1}^{*,-}\theta_{i+1}^{n,-}a_i^{+}\left(\frac{u_{i+1}^{n,-}-a_{i+1}^{-}}{a_{i+1}^{-}-a_i^{+}}\right).$$

According to the third component of (3.1), it follows that

$$(h\theta)_i^{n+1}=(h\theta)_i^{n}-\frac{\Delta t}{\Delta x}\left(\mathbf{F}^{(3)}_{i+\frac{1}{2}}-\mathbf{F}^{(3)}_{i-\frac{1}{2}}\right)=\frac{(h_i^{n,-}+h_i^{n,+})(\theta_i^{n,-}+\theta_i^{n,+})}{4}-\frac{\Delta t}{\Delta x}\left(\mathbf{F}^{(3)}_{i+\frac{1}{2}}-\mathbf{F}^{(3)}_{i-\frac{1}{2}}\right).$$

Then, according to (3.17), yields

$$\begin{aligned}
(h\theta)_i^{n+1} \quad &\geq \quad h_i^{*,+}\theta_i^{n,+}\left[\frac{1}{4}-\frac{\Delta t}{\Delta x}a_{i+1}^{-}\left(\frac{u_i^{n,+}-a_i^{+}}{a_{i+1}^{-}-a_i^{+}}\right)\right] \\
&\quad + h_i^{*,-}\theta_i^{n,-}\left[\frac{1}{4}+\frac{\Delta t}{\Delta x}a_{i-1}^{+}\left(\frac{a_i^{-}-u_i^{n,-}}{a_i^{-}-a_{i-1}^{+}}\right)\right] \\
&\quad - \frac{\Delta t}{\Delta x}h_{i+1}^{*,-}\theta_{i+1}^{n,-}a_i^{+}\left(\frac{a_{i+1}^{-}-u_{i+1}^{n,-}}{a_{i+1}^{-}-a_i^{+}}\right) \\
&\quad + \frac{\Delta t}{\Delta x}h_{i-1}^{*,+}\theta_{i-1}^{n,+}a_i^{-}\left(\frac{u_{i-1}^{n,+}-a_{i-1}^{+}}{a_i^{-}-a_{i-1}^{+}}\right) \\
&\quad + \frac{1}{4}\Big(h_i^{n,-}\theta_i^{n,+}+h_i^{n,+}\theta_i^{n,-}\Big).
\end{aligned} \tag{3.20}$$

We emphasize that the less restrictive condition $\Delta t\leq\dfrac{\Delta x}{2a}$ would be sufficient to ensure the non-negativity of the water depth, as follows from (3.19). However, the estimate for $(h\theta)_i^{n+1}$ in (3.20) contains coefficients whose non-negativity requires $\dfrac{\Delta t}{\Delta x}a\leq\dfrac{1}{4}$. Consequently, the more restrictive CFL condition $\Delta t\leq\dfrac{\Delta x}{4a}$ is needed to guarantee simultaneously the non-negativity of both h and $h\theta$, and hence of the temperature θ.

Assume the water depth and the temperature are nonnegative at time level $t = t^n$. The reconstructions (3.4) and (3.5) then ensure the positivity of $h_i^{n,\pm}$ and $\theta_i^{n,\pm}$ for all i. Consequently, the last term on the right-hand side of (3.20) is nonnegative. In addition, the reconstruction (3.9) guarantees that $h_i^{*,\pm} \geq 0$ for all i. The first four terms on the right-hand side of (3.20) can be written as a linear combination of the quantities $h_i^{*,\pm}\theta_i^{n,\pm}$ and $h_{i\pm1}^{*,\pm}\theta_{i\pm1}^{n,\pm}$, with coefficients analogous to those appearing in (3.19). Therefore, under the CFL condition (3.16), these coefficients are nonnegative. As a result, $(h\theta)_i^{n+1} \geq 0$. If $h_i^{n+1} > 0$, then

$$\theta_i^{n+1} = \frac{(h\theta)_i^{n+1}}{h_i^{n+1}} \geq 0.$$

If $h_i^{n+1} = 0$, the value of θ_i^{n+1} is computed using the desingularization procedure described in Remark 3.1, which preserves $\theta_i^{n+1} \geq 0$. This completes the proof.

□

Remark 3.3. The positivity-preserving property established above concerns the water depth h and the quantity $p = h\theta$ (and hence θ), which are governed by the first and third components of the finite-volume scheme. In the proposed reformulation of the Ripa system (2.2), only the second component of the numerical flux is modified, while the first and third components remain unchanged. As a consequence, the positivity analysis and arguments of [14] carry over with only minor adaptations. The main difference being that the first equation is written in terms of the surface elevation η rather than the water depth h, with $h = \eta - b$ recovered at the discrete level.

Remark 3.4. The positivity of the water depth and the temperature is still valid by using the second-order SSP Runge-Kutta method (3.15), because its solution is a convex combination of two first-order solutions.

3.1.4 Source term discretization

Our objective is to construct a well-balanced numerical scheme that preserves two physically relevant hydrostatic equilibria: the lake-at-rest steady state (1.3), in both wet and dry regions, and the isobaric steady state (1.4). In fully wet configurations, the lake-at-rest solution (1.3) is naturally preserved and does not require any special treatment. However, in the presence of dry regions, this well-balanced property is generally lost.

At the discrete level, we follow the decomposition of the source term introduced in (2.5). The term S_1 is discretized so as to recover the lake-at-rest steady state (1.3), including in the presence of wetting and drying, following the approach of [18]. The term S_2 is discretized consistently with its continuous counterpart and is specifically designed to ensure the preservation of the isobaric steady state (1.4). The resulting discrete source term is defined as

$$\mathbf{S}_i = \begin{pmatrix} 0 \\ S_{1,i}^{(2)} + S_{2,i}^{(2)} \\ 0 \end{pmatrix}. \tag{3.21}$$

The discretization of S_1 is given by

$$S_{1,i}^{(2)} = -g\left(\frac{\eta_i^{*,-} + \eta_i^{*,+}}{2}\right)\left(\frac{\theta_i^{n,-} + \theta_i^{n,+}}{2}\right)\left(\frac{b_{i+1/2}^n - b_{i-1/2}^n}{\Delta x}\right) + S_{b_{i-1/2}} + S_{b_{i+1/2}}, \tag{3.22}$$

where the additional corrective terms accounting for dry regions are defined by

$$S_{b_{i-1/2}} = g\,\Delta_{b_{i-1/2}}\left(\frac{\theta_i^{n,-} + \theta_i^{n,+}}{2}\right)\frac{b_{i+1/2}^n - (b_{i-1/2}^n - \Delta_{b_{i-1/2}})}{2\Delta x}, \tag{3.23}$$

and

$$S_{b_{i+1/2}} = g\,\Delta_{b_{i+1/2}}\left(\frac{\theta_i^{n,-} + \theta_i^{n,+}}{2}\right)\frac{(b_{i+1/2}^n - \Delta_{b_{i+1/2}}) - b_{i-1/2}^n}{2\Delta x}, \tag{3.24}$$

with

$$\Delta_{b_{i-1/2}} = \max\Big(0, -(\eta_i^{n,-} - b_{i-1/2}^n)\Big), \tag{3.25}$$

and

$$\Delta_{b_{i+1/2}} = \max\Big(0, -(\eta_i^{n,+} - b_{i+1/2}^n)\Big)\,. \tag{3.26}$$

In purely wet configurations, the corrective terms $S_{b_{i-1/2}}$ and $S_{b_{i+1/2}}$ vanish and therefore do not affect the scheme.

The discretization of S_2 is given by

$$S_{2,i}^{(2)} = -\frac{g}{2}\left(\frac{b_{i+1/2}^n + b_{i-1/2}^n}{2}\right)^2\left(\frac{\theta_{i+1}^{n,-} + \theta_i^{n,+} - \theta_i^{n,-} - \theta_{i-1}^{n,+}}{2\Delta x}\right). \tag{3.27}$$

This term does not contribute to the balance of the fluxes for the lake-at-rest steady state (1.3), since θ is constant in that configuration, but is essential for ensuring the preservation of the isobaric steady state (1.4), as proved in Theorem 3.7.

Theorem 3.5. *(Well-balancing) The numerical scheme* (3.1)*, combined with the face reconstructions* (3.4)–(3.8)*, the reconstructed states* (3.9)–(3.10)*, the numerical flux* (3.11)*, and the discretized source term* (3.21)*, is well-balanced with respect to the lake-at-rest steady state* (1.3)*, in the presence of wetting and drying.*

Proof. We will consider two typical configurations to prove the well-balancing property in wet-bed and dry-bed simulations. In the first case, we consider a lake-at-rest steady state ($\eta \equiv$ constant, $u \equiv 0$, and $\theta \equiv$ constant) in a wet cell whose two neighboring cells are also wet, see Figure 2a. Because the water is at rest and the velocity vanishes, then $\mathbf{K}_i^{(1)} = 0$ and $\mathbf{K}_i^{(3)} = 0$ are trivial. Based on the aforementioned Riemann states reconstruction (3.6), we have $b_{i-\frac{1}{2}} = b_i^-$ and $b_{i+\frac{1}{2}} = b_{i+1}^-$. Clearly, the stage components of the Riemann states of the water surface elevation η and temperature θ are constant and equal to η_c and θ_c respectively. Notice that, for the still water across the three wet cells, we can obtain $a_{i+1}^- = -a_i^+$ and $a_i^- = -a_{i-1}^+$. The second component of the numerical flux at the cell interface $i + \frac{1}{2}$ is computed by:

$$\mathbf{F}_{i+1/2}^{(2)} = \frac{g}{2}(\eta_c^2 - 2\eta_c b_{i+\frac{1}{2}})\theta_c,$$

which yields

$$\frac{\mathbf{F}_{i+1/2}^{(2)} - \mathbf{F}_{i-1/2}^{(2)}}{\Delta x} = -g\eta_c\theta_c\left(\frac{b_{i+\frac{1}{2}} - b_{i-\frac{1}{2}}}{\Delta x}\right). \tag{3.28}$$

We now compare the discrete flux difference obtained in (3.28) with the discretized source term (3.21). Using (3.25)-(3.26), it follows immediately that $\Delta_{b_{i-1/2}} = 0$ and $\Delta_{b_{i+1/2}} = 0$ and thus $S_{b_{i-1/2}} = 0$ and $S_{b_{i+1/2}} = 0$ for this case. Therefore, the discretization of the second component of the source term (3.21) reduces to:

$$S_{1,i}^{(2)} = -g\eta_c\theta_c\left(\frac{b_{i+\frac{1}{2}} - b_{i-\frac{1}{2}}}{\Delta x}\right) \quad \text{and} \quad S_{2,i}^{(2)} = 0 \text{ (since } \theta = \theta_c \text{ across three cells).}$$

Consequently, it follows that

$$\mathbf{K}_i^{(2)} = \frac{\mathbf{F}_{i+1/2}^{(2)} - \mathbf{F}_{i-1/2}^{(2)}}{\Delta x} - \mathbf{S}_i^{(2)} = 0.$$

This proves that the scheme is well-balanced for the lake-at-rest steady state (1.3) in the fully wet case.

In the second case, we consider a wet cell neighboring a wet cell in the left and a dry cell on the right, see Figure 2b. At the cell interface $i - \frac{1}{2}$, we have $b_{i-\frac{1}{2}} = \max\big(b_{i-1}^+, b_i^-\big) = b_i^-$. The Riemann states satisfy $\eta_{i-1}^{*,+} = \eta_i^{*,-} = \eta_c$. Here, η_c denotes the water surface elevation in the wet-bed area and is constant across all wet cells for a motionless steady state. Notice that, for the still water across the wet cell under consideration and its wet left neighbor, we can obtain $a_i^- = -a_{i-1}^+$. The second component of the numerical flux at the cell interface $i - \frac{1}{2}$ is computed by:

$$\mathbf{F}_{i-1/2}^{(2)} = \frac{g}{2}(\eta_c^2 - 2\eta_c b_{i-\frac{1}{2}})\theta_c,$$

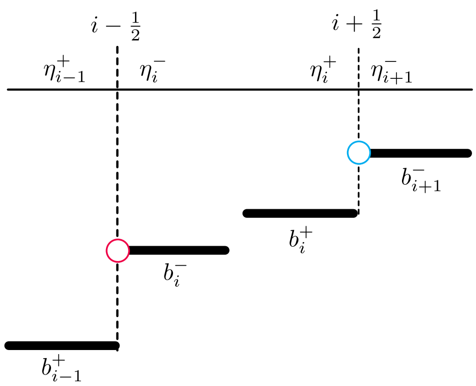
At the cell interface $i + \frac{1}{2}$, we have $b_{i+\frac{1}{2}} = \max\big(b_i^+, b_{i+1}^-\big) = b_{i+1}^-$. After reconstruction (3.9), the stage component of the Riemann states of the water depth take the values of zero, i.e. $h_i^{*,+} = h_{i+1}^{*,-} = 0$. Therefore, $\eta_i^{*,+} = \eta_{i+1}^{*,-} = b_{i+\frac{1}{2}} = \eta_d$ where η_d represents the 'water surface elevation' on the dry ground.

Notice that, for the still water across the wet cell under consideration and its dry right neighbor, we can obtain $a^-_{i+1} = a^+_i = 0$ since $u^+_i = u^-_{i+1} = 0$ and $h^{*,+}_i = h^{*,-}_{i+1} = 0$. The second component of the numerical flux at the cell interface $i + \frac{1}{2}$ is computed by:

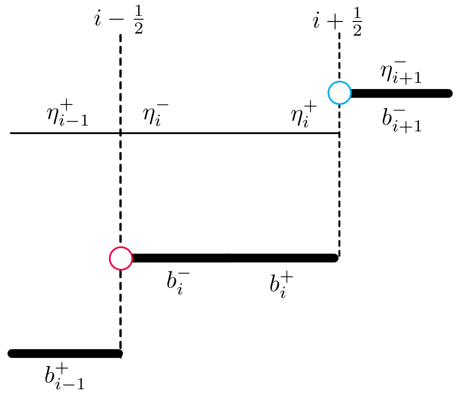
$$\mathbf{F}^{(2)}_{i+1/2} = \frac{g}{2}(\eta_d^2 - 2\eta_d b_{i+\frac{1}{2}})\theta_c = -\frac{g}{2}\eta_d^2\theta_c,$$

which yields

$$\frac{\mathbf{F}^{(2)}_{i+1/2} - \mathbf{F}^{(2)}_{i-1/2}}{\Delta x} = -\frac{g}{2\Delta x}(\eta_d^2 + \eta_c^2 - 2\eta_c b_{i-\frac{1}{2}})\theta_c. \tag{3.29}$$



(a) Wet neighbors



(b) Dry neighbors

Figure 2: Configurations of wet cells with neighboring cells.

We now compare the discrete flux difference obtained in (3.29) with the discretized source term (3.21). Using (3.25), it follows immediately that $\Delta_{b_{i-1/2}} = 0$ and thus $S_{b_{i-1/2}} = 0$. According to (3.26), $\Delta_{b_{i+1/2}} = b_{i+\frac{1}{2}} - \eta^+_i = \eta_d - \eta_c$, hence $S_{b_{i+1/2}} = g(\eta_d - \eta_c)\theta_c\left(\frac{\eta_c - b_{i-1/2}}{2\Delta x}\right)$. Therefore, the discretization of the second component of the source term reduces to

$$S^{(2)}_{1,i} = -g\left(\frac{\eta_c + \eta_d}{2}\right)\theta_c\left(\frac{\eta_d - b_{i-1/2}}{\Delta x}\right) + g(\eta_d - \eta_c)\theta_c\left(\frac{\eta_c - b_{i-1/2}}{2\Delta x}\right) = -\frac{g}{2\Delta x}(\eta_d^2 + \eta_c^2 - 2\eta_c b_{i-\frac{1}{2}})\theta_c,$$

and

$$S^{(2)}_{2,i} = 0 \text{ (since } \theta = \theta_c \text{ across three cells).}$$

Consequently, it follows that

$$\mathbf{K}^{(2)}_i = \frac{\mathbf{F}^{(2)}_{i+1/2} - \mathbf{F}^{(2)}_{i-1/2}}{\Delta x} - \mathbf{S}^{(2)}_i = 0.$$

This proves that the scheme is well-balanced for the lake-at-rest steady state (1.3) in the wet-dry case. □

3.1.5 A numerical method preserving the isobaric steady state

The numerical method proposed in Section 3 exactly preserves the lake-at-rest steady state (1.3). However, it fails to preserve the isobaric steady state (1.4), as illustrated in Figure 8 (see solid red lines). To preserve the isobaric steady state we adopt a positivity-preserving parameter introduced in [14] and an appropriate discretization of the second component of the source term, which is specific to the reformulated Ripa system proposed in this work, see Section 3.1.4. We detail this combined strategy below. We begin by introducing the provable positivity-preserving parameter, defined as

$$\varphi_{i+\frac{1}{2}} := \max\left(\varphi^+_i, \varphi^-_{i+1}\right), \tag{3.30}$$

where

$$\begin{aligned}\varphi_i^+ &= \max\left(\frac{\min\left(u_i^+,0\right)}{a_i^+}, \frac{-\min\left(q_i^{*,+},0\right)}{\max\left(\left|q_i^{*,+}\right|,1\right)}\right),\\ \varphi_{i+1}^- &= \max\left(\frac{\max\left(u_{i+1}^-,0\right)}{a_{i+1}^-}, \frac{\max\left(q_{i+1}^{*,-},0\right)}{\max\left(\left|q_{i+1}^{*,-}\right|,1\right)}\right).\end{aligned} \tag{3.31}$$

Following the arguments in [14], the parameter $\varphi_{i+\frac{1}{2}}$ satisfies

$$0 \leq \varphi_{i+\frac{1}{2}} \leq 1. \tag{3.32}$$

Then, we modify the numerical flux across the cell interface to be

$$\mathbf{F}_{i+1/2}^n = \begin{cases} \mathbf{F}(\mathbf{U}_i^{*,+}, b_{i+1/2}^n) & \text{if } a_i^+ \geq 0,\\ \dfrac{a_{i+1}^-\mathbf{F}(\mathbf{U}_i^{*,+}, b_{i+1/2}^n) - a_i^+\mathbf{F}(\mathbf{U}_{i+1}^{*,-}, b_{i+1/2}^n) + a_{i+1}^- a_i^+ \varphi_{i+\frac{1}{2}}(\mathbf{U}_{i+1}^{*,-} - \mathbf{U}_i^{*,+})}{a_{i+1}^- - a_i^+} & \text{if } a_i^+ \leq 0 \leq a_{i+1}^-,\\ \mathbf{F}(\mathbf{U}_{i+1}^{*,-}, b_{i+1/2}^n) & \text{if } a_{i+1}^- \leq 0,\end{cases} \tag{3.33}$$

with the wave-speed estimates a_i^+ and a_{i+1}^- as defined in (3.12).

The positivity-preserving property remains valid under the modified numerical flux (3.33), since the parameter $\varphi_{i+1/2}$ is defined exactly as in [14] and satisfies (3.32).

Theorem 3.6. *(positivity-preserving) Assume that the system* (3.1) *is discretized using a first-order forward Euler step in time. Assuming $h_i^n \geq 0$, $\theta_i^n \geq 0$ for all $i \in [1, N]$, and the CFL condition satisfies*

$$\Delta t \leq \frac{\Delta x}{4a}, \quad \text{where } a = \max\left(\max_i(a_{i+1}^-, -a_i^+)\right), \tag{3.34}$$

with a_{i+1}^- and a_i^+ defined in (3.12)*, then we obtain $h_i^{n+1} \geq 0$, $\theta_i^{n+1} \geq 0$ for all $i \in [1, N]$ using the numerical scheme* (3.1)*, the modified numerical flux* (3.33) *and* (3.21)*.*

Proof. The proof is a direct adaptation of Theorem 3.7 in [14], since the present scheme employs the same positivity-preserving parameter and wave-speed bounds. We emphasize that the modification introduced in the present formulation affects only the second component of the numerical flux, corresponding to the momentum equation. The first and third components, which govern the evolution of the water depth h and the quantity $h\theta$, respectively, retain the same positivity-preserving HLL structure as in [14]. Moreover, the corresponding first and third components of the source term vanish. Consequently, the modified momentum flux does not enter the updates of either h or $h\theta$, and the key inequalities ensuring the nonnegativity of all coefficients in the corresponding update formulas remain unchanged. Under the CFL condition (3.34), these updates can therefore be written as convex combinations of nonnegative reconstructed states. It follows that $h_i^{n+1} \geq 0$ and $(h\theta)_i^{n+1} \geq 0$, and hence $\theta_i^{n+1} \geq 0$. ☐

In what follows, we prove that the proposed scheme is well-balanced with respect to the isobaric steady state (1.4). To this end, we introduce the equilibrium variables

$$\mathbf{W} = (\eta, q, \Pi)^\top := (\eta, hu, h^2\theta)^\top.$$

Instead of reconstructing the variables $\mathbf{V} = (\eta, q, \theta)^\top$, we reconstruct the equilibrium variables $\mathbf{W}$ using the same MUSCL procedure (3.2). In particular, on each cell interface $x_{i+1/2}$ we define

$$\begin{cases} \mathbf{W}_i^{n,+} = \mathbf{W}_i^n + \dfrac{\Delta x}{2}\partial_x \mathbf{W}_i^n,\\ \mathbf{W}_{i+1}^{n,-} = \mathbf{W}_{i+1}^n - \dfrac{\Delta x}{2}\partial_x \mathbf{W}_{i+1}^n.\end{cases} \tag{3.35}$$

The face values of the water depth, bottom topography and velocity are obtained as in (3.5)-(3.7). The face values of the temperature are then recovered from

$$\begin{cases} \theta_i^{n,+} = \dfrac{\Pi_i^{n,+}}{(h_i^{n,+})^2},\\ \theta_{i+1}^{n,-} = \dfrac{\Pi_{i+1}^{n,-}}{(h_{i+1}^{n,-})^2}.\end{cases} \tag{3.36}$$

This reconstruction is applied in fully wet regions, where $h_i^{n,\pm} > 0$, which is consistent with the isobaric steady state. Finally, using (3.8)-(3.9), the reconstructed (hydrostatic) Riemann states used in the numerical flux are defined by

$$\begin{cases} \eta_i^{*,+} = h_i^{*,+} + b_{i+1/2}^n, & \eta_{i+1}^{*,-} = h_{i+1}^{*,-} + b_{i+1/2}^n, \\ q_i^{*,+} = h_i^{*,+} u_i^{n,+}, & q_{i+1}^{*,-} = h_{i+1}^{*,-} u_{i+1}^{n,-}, \\ \Pi_i^{*,+} = \left(h_i^{*,+}\right)^2 \theta_i^{n,+}, & \Pi_{i+1}^{*,-} = \left(h_{i+1}^{*,-}\right)^2 \theta_{i+1}^{n,-}, \end{cases} \tag{3.37}$$

and

$$p_i^{*,+} = h_i^{*,+} \theta_i^{n,+} = \frac{\Pi_i^{*,+}}{h_i^{*,+}}, \qquad p_{i+1}^{*,-} = h_{i+1}^{*,-} \theta_{i+1}^{n,-} = \frac{\Pi_{i+1}^{*,-}}{h_{i+1}^{*,-}}. \tag{3.38}$$

Theorem 3.7. *(Well-balancing) The numerical scheme* (3.1)*, combined with the MUSCL reconstruction procedure* (3.2)–(3.35) *applied to the equilibrium variables* $\mathbf{W} = (\eta, q, \Pi)$ *with* $\Pi = h^2\theta$*, the geometric reconstructions* (3.5)–(3.8) *and the hydrostatic reconstruction* (3.9)*, the reconstructed Riemann states* (3.37)–(3.38)*, the modified numerical flux* (3.33)*, and the discretized source term* (3.21)*, is well-balanced with respect to the isobaric steady state* (1.4)*.*

Proof. For the isobaric steady state (1.4), the velocity vanishes and the bottom topography is constant ($b = b_0$, where b_0 is a given value). Since the velocity vanishes, according to the definition of the parameter in (3.30)-(3.31), we can obtain

$$\varphi_{i+\frac{1}{2}} = 0, \quad q_i^{*,+} = q_{i+1}^{*,-} = 0, \quad a_i^+ = -a_{i+1}^-.$$

Moreover, if the bottom topography is constant, i.e. $b(x) \equiv b_0$, then the MUSCL reconstruction procedure applied to the equilibrium variables $\mathbf{W}$, together with (3.5)–(3.6), preserves this property. Indeed, since $\eta_i^n = h_i^n + b_0$ for all i, the corresponding slopes satisfy $\psi_i^n(\eta) = \psi_i^n(h)$, which implies that the reconstructed bottom topography at cell interfaces remains constant:

$$b_i^{n,+} = \eta_i^{n,+} - h_i^{n,+} = b_0, \qquad b_{i+1}^{n,-} = \eta_{i+1}^{n,-} - h_{i+1}^{n,-} = b_0.$$

Moreover, using (3.8), the bottom topography at the cell interface is defined by

$$b_{i+1/2}^n := \max\left(b_i^{n,+},\, b_{i+1}^{n,-}\right) = b_0.$$

As a result, the reconstruction does not introduce any spurious variations of the bottom topography and is fully consistent with the preservation of the isobaric steady-state solution (1.4).

We now verify that, under the isobaric steady state (1.4), the discrete flux difference exactly balances the discretized source term introduced in Section 3.1.4. According to the definition of the fluxes (3.33), one has

$$\frac{a_{i+1}^- a_i^+ \varphi_{i+\frac{1}{2}} (\mathbf{U}_{i+1}^{*,-} - \mathbf{U}_i^{*,+})}{a_{i+1}^- - a_i^+} = 0.$$

Then, it follows immediately that

$$\mathbf{K}_i^{(1)} = 0 \quad \text{and} \quad \mathbf{K}_i^{(3)} = 0.$$

The second component of the numerical flux at the cell interface $i + \frac{1}{2}$ is

$$\mathbf{F}_{i+1/2}^{(2)} = \frac{g}{4}\Big[\big(\eta_i^{*,+^2} - 2\eta_i^{*,+} b_0\big)\theta_i^{n,+} + \big(\eta_{i+1}^{*,-^2} - 2\eta_{i+1}^{*,-} b_0\big)\theta_{i+1}^{n,-}\Big]. \tag{3.39}$$

Using (3.39), we can obtain the difference of the numerical flux function as

$$\begin{aligned} \mathbf{F}_{i+1/2}^{(2)} - \mathbf{F}_{i-1/2}^{(2)} \;=\; & \frac{g}{4}\Big[\big(\eta_i^{*,+^2} - 2\eta_i^{*,+} b_0\big)\theta_i^{n,+} + \big(\eta_{i+1}^{*,-^2} - 2\eta_{i+1}^{*,-} b_0\big)\theta_{i+1}^{n,-}\Big] \\ & - \frac{g}{4}\Big[\big(\eta_{i-1}^{*,+^2} - 2\eta_{i-1}^{*,+} b_0\big)\theta_{i-1}^{n,+} + \big(\eta_i^{*,-^2} - 2\eta_i^{*,-} b_0\big)\theta_i^{n,-}\Big]. \end{aligned} \tag{3.40}$$

Thanks to the fact that, in the isobaric steady state (1.4), one has

$$\frac{g}{2}\Pi = \frac{g}{2}h^2\theta = \text{constant},$$

and since the MUSCL reconstruction is applied directly to the equilibrium variable Π, this constancy is preserved at the reconstructed states. Consequently, the following discrete identity holds:

$$\begin{aligned}
&\frac{g}{4}\Big[\big({\eta_i^{*,+}}^2 - 2\eta_i^{*,+}b_0 + b_0^2\big)\theta_i^{n,+} + \big({\eta_{i+1}^{*,-}}^2 - 2\eta_{i+1}^{*,-}b_0 + b_0^2\big)\theta_{i+1}^{n,-}\Big] \\
&= \frac{g}{4}\Big[\big({\eta_{i-1}^{*,+}}^2 - 2\eta_{i-1}^{*,+}b_0 + b_0^2\big)\theta_{i-1}^{n,+} + \big({\eta_i^{*,-}}^2 - 2\eta_i^{*,-}b_0 + b_0^2\big)\theta_i^{n,-}\Big].
\end{aligned} \tag{3.41}$$

Using (3.41), the difference of the numerical fluxes (3.40) can therefore be rewritten as

$$\mathbf{F}^{(2)}_{i+1/2} - \mathbf{F}^{(2)}_{i-1/2} = -\frac{g}{4}\, b_0^2\big(\theta_{i+1}^{n,-} + \theta_i^{n,+} - \theta_i^{n,-} - \theta_{i-1}^{n,+}\big). \tag{3.42}$$

We now compare the discrete flux difference obtained in (3.42) with the discretized source term (3.21). Using (3.25)-(3.26), it follows immediately that $\Delta_{b_{i-1/2}} = 0$ and $\Delta_{b_{i+1/2}} = 0$ and thus $S_{b_{i-1/2}} = 0$ and $S_{b_{i+1/2}} = 0$ for this case. Therefore, the discretization of the second component of the source term (3.21) reduces to:

$$S_{1,i}^{(2)} = 0 \text{ (since } b \text{ is constant)} \quad \text{and} \quad S_{2,i}^{(2)} = -\frac{g}{2}\left(\frac{b^n_{i+1/2} + b^n_{i-1/2}}{2}\right)^2 \left(\frac{\theta_{i+1}^{n,-} + \theta_i^{n,+} - \theta_i^{n,-} - \theta_{i-1}^{n,+}}{2\Delta x}\right).$$

Consequently, it follows that

$$\mathbf{K}_i^{(2)} = \frac{\mathbf{F}^{(2)}_{i+1/2} - \mathbf{F}^{(2)}_{i-1/2}}{\Delta x} - \mathbf{S}_i^{(2)} = 0.$$

This proves that the scheme is well-balanced for the isobaric steady state (1.4). □

Remark 3.8. The well-balanced property associated with (1.3) is still valid when the numerical scheme is equipped with the modified numerical flux (3.33), since the parameter is zero when the water is at rest.

Algorithmic summary. For completeness, the main steps of the proposed one-dimensional scheme are summarized below. At each stage of the SSP-RK2 method, the following operations are performed:

1. **MUSCL reconstruction.** Reconstruct the appropriate variables at the cell interfaces. The variables

   $$\mathbf{V} = (\eta, q, \theta)^\top$$

   are used for the lake-at-rest steady state, whereas the equilibrium variables

   $$\mathbf{W} = (\eta, q, \Pi)^\top, \qquad \Pi = h^2\theta,$$

   are used for the isobaric steady state.

2. **Hydrostatic reconstruction.** Compute the interface bottom elevations according to (3.8) and reconstruct the nonnegative water depths and the corresponding Riemann states according to (3.9)-(3.10).

3. **Numerical flux.** Evaluate the modified HLL numerical flux using the reconstructed Riemann states, the wave-speed estimates, and the positivity-preserving parameter $\phi_{i+\frac{1}{2}}$.

4. **Source-term discretization.** Compute the discrete source term, including the corrections required near wet-dry interfaces and the temperature-dependent contribution required for preserving the isobaric steady state.

5. **Time update.** Form the finite-volume spatial operator and advance the solution using the SSP-RK2 method. The reconstruction, numerical fluxes, and source terms are recomputed at the intermediate Runge–Kutta stage.

3.2 Two-dimensional scheme

In this section, we extend the one-dimensional well-balanced, positivity-preserving finite-volume scheme to the two-dimensional Ripa system. Following the methodology developed in the one-dimensional case, we first derive the corresponding reformulation of the Ripa system in two spatial dimensions. The resulting two-dimensional formulation can be written as

$$\begin{cases} \partial_t \eta + \partial_x(hu) + \partial_y(hv) = 0 \ , \\ \partial_t(hu) + \partial_x(hu^2) + \dfrac{g}{2}\partial_x\Big((\eta^2 - 2\eta b)\theta\Big) + \partial_y(huv) = -g\eta\theta\partial_x b - \dfrac{g}{2}b^2\partial_x\theta \ , \\ \partial_t(hv) + \partial_x(huv) + \partial_y(hv^2) + \dfrac{g}{2}\partial_y\Big((\eta^2 - 2\eta b)\theta\Big) = -g\eta\theta\partial_y b - \dfrac{g}{2}b^2\partial_y\theta \ , \\ \partial_t(h\theta) + \partial_x(hu\theta) + \partial_y(hv\theta) = 0. \end{cases} \tag{3.43}$$

where $v(x, y, t)$ represents the velocity in the $y-$direction.

In matrix form the Ripa system (3.43) can be expressed as:

$$\partial_t \mathbf{U} + \partial_x \mathbf{F}(\mathbf{U}, b) + \partial_y \mathbf{G}(\mathbf{U}, b) = \mathbf{S}(\mathbf{U}, b), \tag{3.44}$$

with

$$\mathbf{U} = \begin{pmatrix} \eta \\ q_x \\ q_y \\ p \end{pmatrix} = \begin{pmatrix} \eta \\ hu \\ hv \\ h\theta \end{pmatrix}, \quad \mathbf{F}(\mathbf{U}, b) = \begin{pmatrix} q_x \\ \dfrac{q_x^2}{h} + \dfrac{g}{2}(\eta^2 - 2\eta b)\dfrac{p}{h} \\ \dfrac{q_x q_y}{h} \\ \dfrac{q_x p}{h} \end{pmatrix}, \quad \mathbf{G}(\mathbf{U}, b) = \begin{pmatrix} q_y \\ \dfrac{q_x q_y}{h} \\ \dfrac{q_y^2}{h} + \dfrac{g}{2}(\eta^2 - 2\eta b)\dfrac{p}{h} \\ \dfrac{q_y p}{h} \end{pmatrix}, \tag{3.45}$$

and

$$\mathbf{S}(\mathbf{U}, b) = \begin{pmatrix} 0 \\ -g\eta\dfrac{p}{h}\partial_x b - \dfrac{g}{2}b^2\partial_x\Big(\dfrac{p}{h}\Big) \\ -g\eta\dfrac{p}{h}\partial_y b - \dfrac{g}{2}b^2\partial_y\Big(\dfrac{p}{h}\Big) \\ 0 \end{pmatrix}. \tag{3.46}$$

Spatial discretization. All components of the scheme are extended from the one-dimensional case in a dimension-by-dimension manner. We thus restrict ourselves to an algorithmic description of the two-dimensional scheme. We begin by discretizing system (3.43) in two spatial dimensions. Let $\Omega = [0, L_x] \times [0, L_y]$ denote the computational domain, and let $N_x, N_y \in \mathbb{N}^*$ be the number of cells in the x- and y-directions, respectively. The domain is partitioned into rectangular control volumes

$$m_{i,j} = (x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}) \times (y_{j-\frac{1}{2}}, y_{j+\frac{1}{2}}), \qquad 1 \le i \le N_x, \ 1 \le j \le N_y.$$

The cell centers are defined by

$$x_i = \frac{x_{i-\frac{1}{2}} + x_{i+\frac{1}{2}}}{2}, \qquad y_j = \frac{y_{j-\frac{1}{2}} + y_{j+\frac{1}{2}}}{2},$$

and the uniform mesh sizes by

$$\Delta x = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}, \qquad \Delta y = y_{j+\frac{1}{2}} - y_{j-\frac{1}{2}}.$$

The grid interfaces $x_{1/2}$ and $x_{N_x+1/2}$ (resp. $y_{1/2}$ and $y_{N_y+1/2}$) correspond to the left and right (resp. bottom and top) boundaries of the computational domain (see Figure 3).

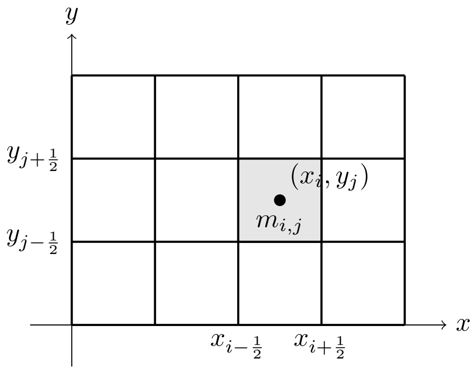


Figure 3: Two-dimensional Cartesian discretization of the computational domain. The control volume $m_{i,j}$ is bounded by the interfaces $x_{i\pm\frac{1}{2}}$ and $y_{j\pm\frac{1}{2}}$, and (x_i, y_j) denotes the cell center.

For each control volume $m_{i,j}$, the cell averages $\mathbf{U}_{i,j}^n = (\eta, q_x, q_y, p)_{i,j}^n$ are updated according to the finite-volume formula

$$\mathbf{U}_{i,j}^{n+1} = \mathbf{U}_{i,j}^n - \Delta t\left(\frac{\mathbf{F}_{i+\frac{1}{2},j}^n - \mathbf{F}_{i-\frac{1}{2},j}^n}{\Delta x} + \frac{\mathbf{G}_{i,j+\frac{1}{2}}^n - \mathbf{G}_{i,j-\frac{1}{2}}^n}{\Delta y}\right) + \Delta t\, \mathbf{S}_{i,j}^n, \tag{3.47}$$

where the numerical fluxes $\mathbf{F}_{i+\frac{1}{2},j}^n$ and $\mathbf{G}_{i,j+\frac{1}{2}}^n$ are computed using a one-dimensional HLL approximate Riemann solver applied in the normal direction to each interface and $\mathbf{S}_{i,j}^n$ is the discretized source term.

Reconstruction. The MUSCL reconstruction with the generalized minmod limiter is applied independently in the x- and y-directions. As in the one-dimensional case, the reconstruction is performed either on the primitive variables (η, q_x, q_y, θ) or on the equilibrium variables (η, q_x, q_y, Π), depending on whether the lake-at-rest or the isobaric steady state is targeted. Hydrostatic reconstruction is then used at all interfaces to guarantee the positivity of the water depth and temperature in the presence of wetting and drying. The superscripts $(\cdot)^+$ and $(\cdot)^-$ denote the hydrostatically reconstructed states on the two sides of the interface, corresponding to the adjacent cells sharing that interface. More precisely, at a vertical interface $x_{i+\frac{1}{2}}$ they represent the states in cells (i, j) and $(i+1, j)$, respectively, while at a horizontal interface $y_{j+\frac{1}{2}}$ they correspond to the states in cells (i, j) and $(i, j+1)$. For the sake of brevity, and to avoid unnecessary repetition, the detailed reconstruction formulas are not reproduced here.

Numerical fluxes. At a vertical interface $x_{i+\frac{1}{2}}$, the HLL numerical flux in the x-direction is defined by

$$\mathbf{F}_{i+\frac{1}{2},j}^n = \begin{cases} \mathbf{F}(\mathbf{U}_{i,j}^{*,+},\, b_{i+\frac{1}{2},j}^n) & \text{if } a_{i+\frac{1}{2},j}^{x,+} \geq 0, \\ \dfrac{a_{i+\frac{1}{2},j}^{x,-}\mathbf{F}(\mathbf{U}_{i,j}^{*,+},\, b_{i+\frac{1}{2},j}^n) - a_{i+\frac{1}{2},j}^{x,+}\mathbf{F}(\mathbf{U}_{i+1,j}^{*,-},\, b_{i+\frac{1}{2},j}^n) + a_{i+\frac{1}{2},j}^{x,-}a_{i+\frac{1}{2},j}^{x,+}\left(\mathbf{U}_{i+1,j}^{*,-} - \mathbf{U}_{i,j}^{*,+}\right)}{a_{i+\frac{1}{2},j}^{x,-} - a_{i+\frac{1}{2},j}^{x,+}} & \text{if } a_{i+\frac{1}{2},j}^{x,+} \leq 0 \leq a_{i+\frac{1}{2},j}^{x,-}, \\ \mathbf{F}(\mathbf{U}_{i+1,j}^{*,-},\, b_{i+\frac{1}{2},j}^n) & \text{if } a_{i+\frac{1}{2},j}^{x,-} \leq 0. \end{cases} \tag{3.48}$$

Similarly, at a horizontal interface $y_{j+\frac{1}{2}}$, the HLL numerical flux in the y-direction is given by

$$\mathbf{G}_{i,j+\frac{1}{2}}^n = \begin{cases} \mathbf{G}(\mathbf{U}_{i,j}^{*,+},\, b_{i,j+\frac{1}{2}}^n) & \text{if } a_{i,j+\frac{1}{2}}^{y,+} \geq 0, \\ \dfrac{a_{i,j+\frac{1}{2}}^{y,-}\mathbf{G}(\mathbf{U}_{i,j}^{*,+},\, b_{i,j+\frac{1}{2}}^n) - a_{i,j+\frac{1}{2}}^{y,+}\mathbf{G}(\mathbf{U}_{i,j+1}^{*,-},\, b_{i,j+\frac{1}{2}}^n) + a_{i,j+\frac{1}{2}}^{y,-}a_{i,j+\frac{1}{2}}^{y,+}\left(\mathbf{U}_{i,j+1}^{*,-} - \mathbf{U}_{i,j}^{*,+}\right)}{a_{i,j+\frac{1}{2}}^{y,-} - a_{i,j+\frac{1}{2}}^{y,+}} & \text{if } a_{i,j+\frac{1}{2}}^{y,+} \leq 0 \leq a_{i,j+\frac{1}{2}}^{y,-}, \\ \mathbf{G}(\mathbf{U}_{i,j+1}^{*,-},\, b_{i,j+\frac{1}{2}}^n) & \text{if } a_{i,j+\frac{1}{2}}^{y,-} \leq 0. \end{cases} \tag{3.49}$$

The corresponding wave-speed estimates are defined in a direction-by-direction manner. At the vertical interface $x_{i+\frac{1}{2}}$, they are given by

$$a_{i+\frac{1}{2},j}^{x,-} = \max\left(u_{i,j}^+ + \sqrt{g\, h_{i,j}^{*,+}\theta_{i,j}^+},\; u_{i+1,j}^- + \sqrt{g\, h_{i+1,j}^{*,-}\theta_{i+1,j}^-},\; 0\right),$$

$$a^{x,+}_{i+\frac{1}{2},j} = \min\left(u^+_{i,j} - \sqrt{g\, h^{*,+}_{i,j}\theta^+_{i,j}},\ u^-_{i+1,j} - \sqrt{g\, h^{*,-}_{i+1,j}\theta^-_{i+1,j}},\ 0 \right),$$

Similarly, at the horizontal interface $y_{j+\frac{1}{2}}$, the wave-speed estimates are defined by

$$a^{y,-}_{i,j+\frac{1}{2}} = \max\left(v^+_{i,j} + \sqrt{g\, h^{*,+}_{i,j}\theta^+_{i,j}},\ v^-_{i,j+1} + \sqrt{g\, h^{*,-}_{i,j+1}\theta^-_{i,j+1}},\ 0 \right),$$

$$a^{y,+}_{i,j+\frac{1}{2}} = \min\left(v^+_{i,j} - \sqrt{g\, h^{*,+}_{i,j}\theta^+_{i,j}},\ v^-_{i,j+1} - \sqrt{g\, h^{*,-}_{i,j+1}\theta^-_{i,j+1}},\ 0 \right).$$

When the isobaric steady-state preservation is targeted, the reconstruction is performed on the equilibrium variables (η, q_x, q_y, Π). The HLL numerical fluxes in two dimensions are then modified following exactly the one-dimensional construction (3.33), applied in a dimension-by-dimension manner. More precisely, at each interface, the intermediate-wave contribution in the HLL flux is multiplied by a positivity-preserving parameter φ, constructed by the same principle as in the one-dimensional case and evaluated using the corresponding normal velocity. At a vertical interface $x_{i+\frac{1}{2}}$, we set

$$\varphi_{i+\frac{1}{2},j} := \max\left(\varphi^{x,+}_{i,j},\ \varphi^{x,-}_{i+1,j}\right), \tag{3.50}$$

with

$$\begin{aligned} \varphi^{x,+}_{i,j} &= \max\left(\frac{\min\left(u^+_{i,j},\ 0\right)}{a^{x,+}_{i+\frac{1}{2},j}},\ \frac{-\min\left(q^{*,+}_{x,i,j},\ 0\right)}{\max\left(\left|q^{*,+}_{x,i,j}\right|,\ 1\right)} \right), \\ \varphi^{x,-}_{i+1,j} &= \max\left(\frac{\max\left(u^-_{i+1,j},\ 0\right)}{a^{x,-}_{i+\frac{1}{2},j}},\ \frac{\max\left(q^{*,-}_{x,i+1,j},\ 0\right)}{\max\left(\left|q^{*,-}_{x,i+1,j}\right|,\ 1\right)} \right). \end{aligned} \tag{3.51}$$

At a horizontal interface $y_{j+\frac{1}{2}}$, the parameter is defined as

$$\varphi_{i,j+\frac{1}{2}} := \max\left(\varphi^{y,+}_{i,j},\ \varphi^{y,-}_{i,j+1}\right), \tag{3.52}$$

with

$$\begin{aligned} \varphi^{y,+}_{i,j} &= \max\left(\frac{\min\left(v^+_{i,j},\ 0\right)}{a^{y,+}_{i,j+\frac{1}{2}}},\ \frac{-\min\left(q^{*,+}_{y,i,j},\ 0\right)}{\max\left(\left|q^{*,+}_{y,i,j}\right|,\ 1\right)} \right), \\ \varphi^{y,-}_{i,j+1} &= \max\left(\frac{\max\left(v^-_{i,j+1},\ 0\right)}{a^{y,-}_{i,j+\frac{1}{2}}},\ \frac{\max\left(q^{*,-}_{y,i,j+1},\ 0\right)}{\max\left(\left|q^{*,-}_{y,i,j+1}\right|,\ 1\right)} \right). \end{aligned} \tag{3.53}$$

Source term discretization. The source term is discretized by splitting its contributions in the x- and y-directions and applying the one-dimensional well-balanced discretizations along grid lines parallel to each coordinate axis. This construction ensures the exact preservation of both the lake-at-rest steady state, including wet-dry interfaces, and the isobaric steady state.

Time discretization and CFL condition. Time integration is performed using the second-order strong stability preserving Runge-Kutta method (SSP-RK2). The time step Δt is chosen to satisfy a two-dimensional CFL condition ensuring the positivity-preserving property of the scheme. More precisely, following the analysis of [4], Δt is required to satisfy

$$\Delta t \le \min\left(\frac{\Delta x}{8a},\ \frac{\Delta y}{8b} \right),$$

where

$$a := \max_{i,j}\left(\max\left(a^{x,-}_{i+\frac{1}{2},j},\ -a^{x,+}_{i+\frac{1}{2},j}\right) \right), \qquad b := \max_{i,j}\left(\max\left(a^{y,-}_{i,j+\frac{1}{2}},\ -a^{y,+}_{i,j+\frac{1}{2}}\right) \right),$$

denote the maximum characteristic wave speeds in the x- and y-directions, respectively.

Desingularization. As in the one-dimensional case, the temperature θ is recovered from the conservative variables at the cell centers using a desingularized division in order to avoid numerical issues when the water depth becomes small. More precisely, for each cell (i,j), $\theta_{i,j}^n$ is computed from $(h_{i,j}^n, p_{i,j}^n)$ using the same desingularization procedure as in Remark 3.1, with $\varepsilon = \max\{\Delta x, \Delta y\}$. In two space dimensions, the maximum of the two mesh spacings is used as a threshold, particularly for nonuniform spatial resolutions in the two coordinate directions. However, in all the two-dimensional numerical experiments considered in this work, the same spatial resolution is used in both coordinate directions, that is, $\Delta x = \Delta y$.

The resulting two-dimensional scheme preserves the lake-at-rest steady state, including wet-dry interfaces, and the isobaric steady state, while maintaining non-negativity of the water depth and temperature.

4 Numerical tests

In this section, we assess the performance of the proposed scheme through a series of one- and two-dimensional numerical tests, including both wet and dry-bed simulations. In all experiments, we take the gravitational constant $g = 1$ and the minmod parameter $\gamma = 1.3$.

4.1 One-dimensional numerical tests

4.1.1 Accuracy test

We first examine the accuracy of the proposed scheme through a smooth convergence test. Following [8, 29, 22], the bottom topography is defined as a sinusoidal hump,

$$b(x) = \sin^2(\pi x), \qquad x \in (0,1). \tag{4.1}$$

All variables are subject to periodic boundary conditions. The initial data are prescribed as

$$h(x,0) = 5 + e^{\sin(2\pi x)}, \qquad hu(x,0) = \sin(\cos(2\pi x)), \qquad \theta(x,0) = \sin(2\pi x) + 2. \tag{4.2}$$

The computation is performed up to time $t = 0.04$, over which the solution remains smooth. A highly resolved reference solution is obtained on a mesh with 12,800 cells. The L^1 errors are evaluated for the variables h, hu, and $h\theta$. The corresponding convergence rates, reported in Table 1, consistently confirm the second-order accuracy of the method.

Table 1: L^1-errors in h, hu and $h\theta$ along with experimental orders of convergence (EOC) at time $t = 0.04$.

N	L^1-error in h	EOC	L^1-error in hu	EOC	L^1-error in $h\theta$	EOC
50	8.921e-04	-	0.0055	-	0.0022	-
100	2.158e-04	2.05	0.0013	2.03	5.640e-04	1.95
200	5.465e-05	1.98	3.269e-04	2.04	1.428e-04	1.98
400	1.294e-05	2.08	7.755e-05	2.07	3.506e-05	2.02
800	3.033e-06	2.09	1.838e-05	2.07	8.432e-06	2.05
1600	7.092e-07	2.09	4.317e-06	2.09	1.973e-06	2.09
3200	1.636e-07	2.11	9.773e-07	2.14	4.329e-07	2.18

We next investigate the convergence properties of the scheme in the presence of a wet-dry interface. The bottom topography is again chosen as

$$b(x) = \sin^2(\pi x), \qquad x \in (0,1). \tag{4.3}$$

Periodic boundary conditions are imposed, and the initial conditions are given by

$$h(x,0) = e^{\cos(2\pi x)} - e^{-1} + 10^{-4}, \qquad u(x,0) = \sin(\cos(2\pi x)), \qquad \theta(x,0) = e^{\cos(2\pi x)}. \tag{4.4}$$

The minimum water depth is therefore 10^{-4}. The simulation is run until $t = 0.04$, during which the solution remains smooth. As before, the reference solution is computed on a refined mesh of 12,800 cells. The L^1 error norms for h, hu, and $h\theta$ are presented in Table 2, where the observed convergence rates remain close to second order despite the presence of a wet-dry front.

Table 2: L^1-errors in h, hu and $h\theta$ along with experimental orders of convergence (EOC) at time $t = 0.04$.

N	L^1-error in h	EOC	L^1-error in hu	EOC	L^1-error in $h\theta$	EOC
50	0.0055	-	0.023	-	0.0052	-
100	0.0012	2.16	0.0058	2.02	0.0013	1.94
200	3.031e-04	2.02	0.0014	2.04	3.452e-04	1.97
400	7.385e-05	2.03	3.350e-04	2.07	8.370e-05	2.04
800	1.844e-05	2.00	8.032e-05	2.06	2.016e-05	2.05
1600	4.513e-06	2.03	1.881e-05	2.09	4.713e-06	2.09
3200	1.036e-06	2.12	4.137e-06	2.18	1.039e-06	2.18

4.1.2 Lake-at-rest steady state with wet-dry interface

In this test case, we assess the well-balanced and positivity-preserving properties of the proposed scheme in both wet-bed and dry-bed configurations. The computational domain is $[-1, 1]$, and we consider a smooth bottom topography defined as in [28] by

$$b(x) = \begin{cases} 2\big(\cos\big(10\pi(x+0.3)\big)+1\big), & -0.4 \le x \le -0.2, \\ \cos\big(10\pi(x-0.3)\big)+1, & 0.2 \le x \le 0.4, \\ 0, & \text{otherwise.} \end{cases}$$

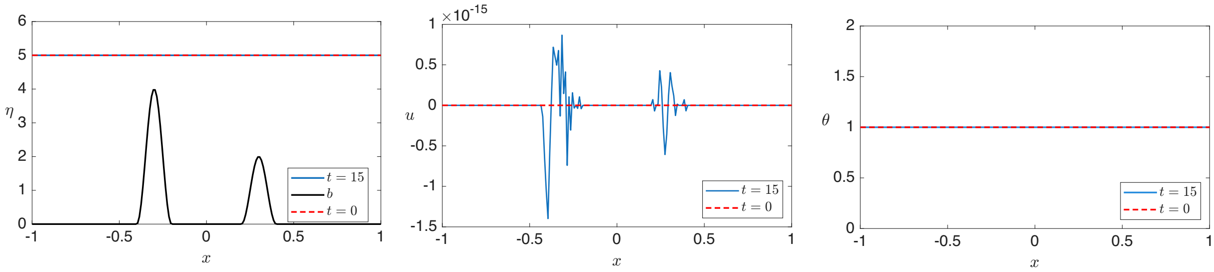
For the wet-bed configuration, the initial conditions are prescribed as

$$\eta(x,0) = \max(5, b(x)), \qquad u(x,0) = 0, \qquad \theta(x,0) = 1,$$

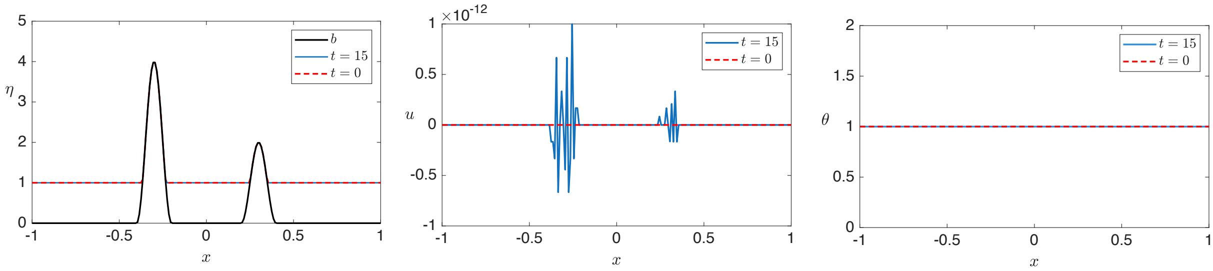
while for the dry-bed configuration they are given by

$$\eta(x,0) = \max(1, b(x)), \qquad u(x,0) = 0, \qquad \theta(x,0) = 1.$$

All simulations are conducted up to the final time $t = 15$ using a uniform grid with $N = 200$ cells with periodic boundary conditions. Figures 4a–4b display the water depth, velocity, and temperature profiles at the initial time $t = 0$ and at $t = 15$, demonstrating that the proposed scheme exactly preserves the steady-state solution in both wet-bed and dry-bed cases. The L^1 and L^∞ errors for the variables η, u,



(a) Wet case



(b) Dry case

Figure 4: Lake-at-rest steady state: The water surface elevation, the velocity, the temperature plots at $t = 15$

and θ are reported in Table 3 for both wet-bed and dry-bed configurations. All errors are either exactly zero or of the order of machine precision, confirming that the proposed scheme preserves the lake-at-rest steady state (1.3) up to round-off accuracy in both wet and dry regimes.

Table 3: Errors for the lake-at-rest steady-state solutions in wet and dry cases.

	L^1 error			L^∞ error		
State	η	u	θ	η	u	θ
Wet	0	1.267e-16	0	0	1.397e-15	0
Dry	0	1.336e-14	0	0	5.322e-13	0

4.1.3 Isobaric steady state

In this test case, we examine the ability of the proposed scheme to preserve the isobaric steady state (1.4). We take the initial data corresponding to the isobaric steady state given in [28] as follows:

$$b(x) = 1, \quad h(x,0) = 1 + \frac{1}{\sqrt{3\pi}} \exp\Big(-\frac{50(x-1.5)^2}{3}\Big), \quad \theta(x,0) = \frac{1}{h(x,0)^2}.$$

The computational domain $[0,3]$ is discretized using a uniform grid with $N = 200$ cells. The simulation is run up to a final time $t = 20$ with periodic boundary conditions. In Table 4, we present the L^1 and L^∞ errors in the water surface, the velocity and the temperature. From the data, we note that the present scheme maintains the isobaric steady state (1.4) up to machine precision.

Table 4: Errors for the isobaric steady-state solution.

	L^1-error			L^∞-error	
η	u	θ	η	u	θ
5.256e-14	1.332e-13	9.170e-14	6.261e-14	6.245e-14	1.104e-13

4.1.4 Dam-break over a non-flat bottom

In this test case, we focus on the dam break problem over a non-flat topography. This test case was also examined in [4, 7, 14, 22, 28]. The bottom topography function is defined as follows:

$$b(x) = \begin{cases} 2\big(\cos\big(10\pi(x+0.3)\big)+1\big), & -0.4 \le x \le -0.2, \\ 0.5\big(\cos\big(10\pi(x-0.3)\big)+1\big), & 0.2 \le x \le 0.4, \\ 0, & \text{otherwise.} \end{cases}$$

We use the following initial data:

$$(\eta, u, \theta)(x,0) = \begin{cases} (5,0,1), & \text{if } x < 0, \\ (1,0,5), & \text{if } x > 0. \end{cases}$$

The solution is computed over the computational domain $[-1,1]$ with 200 uniform cells and transmissive boundary conditions. The reference solutions are computed using 2000 uniform cells. The simulation is run until the final time $t = 0.3$. Initially, the area near $x = 0.3$ is almost dry as the water height is $h(0,0.3) = 0$. In Figure 5, we display the results for the free surface η, the velocity u, the temperature θ and the pressure $\frac{g}{2}\Pi = \frac{g}{2}h^2\theta$. As expected the scheme preserves the positivity of h and θ in the almost dry region. Moreover, the coarse mesh results match well with the fine mesh results.

4.1.5 The perturbed lake-at-rest steady-state flow

We consider this test case to assess the ability of the proposed scheme to accurately capture small perturbations in both the free-surface elevation and the temperature field of an initially lake at rest, especially in the presence of wet–dry interfaces over a variable bottom topography. This benchmark problem has been widely studied in the literature, see for instance [4, 22]. The bottom topography featuring two bumps within the computational domain $[-4,2]$ is defined as follows:

$$b(x) = \begin{cases} 0.85\big(\cos\big(10\pi(x+0.9)\big)+1\big), & -1 \le x \le -0.8, \\ 1.25\big(\cos\big(10\pi(x-0.4)\big)+1\big), & 0.3 \le x \le 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

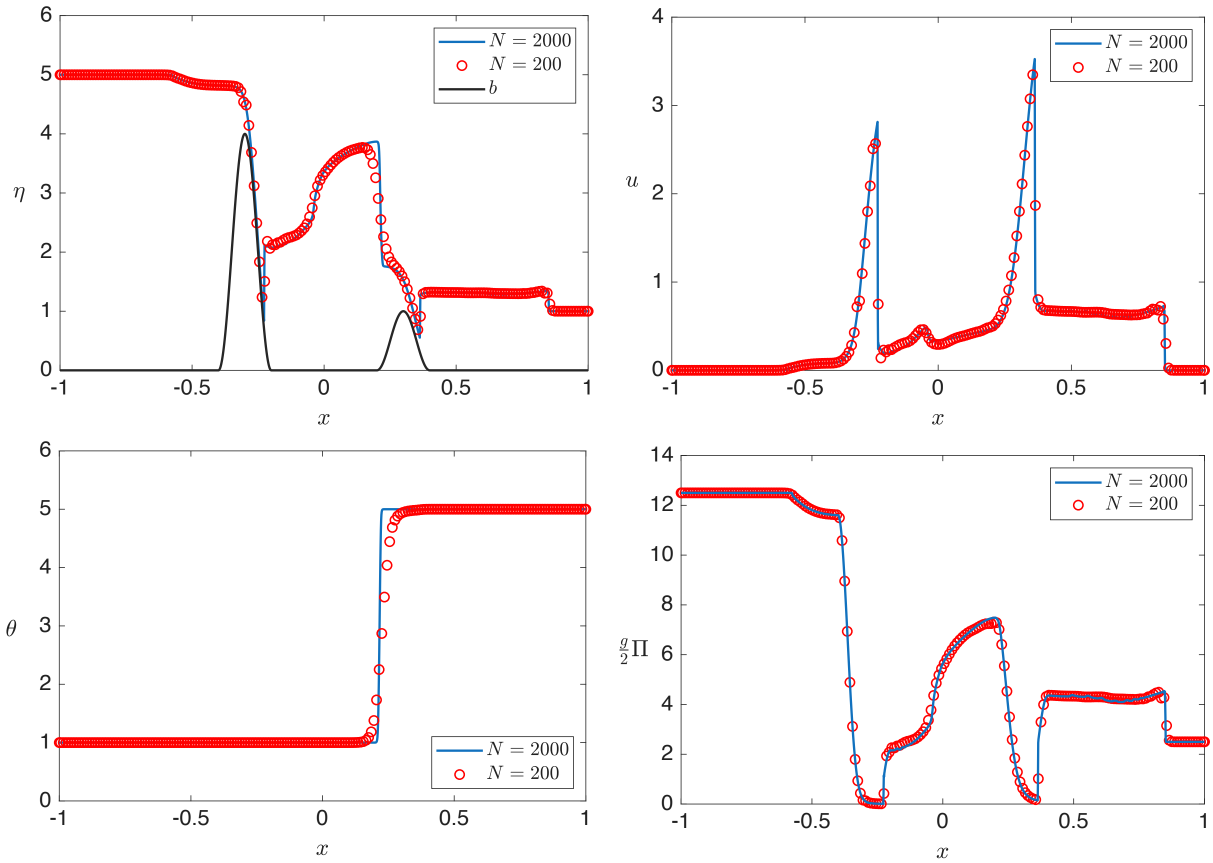


Figure 5: Dam-break over a non-flat bottom. Top left: Water surface elevation. Top right: Velocity. Bottom left: Temperature. Bottom right: Pressure.

We perturb the still-water steady state by introducing a small disturbance of amplitude $\varepsilon = 0.01$ to the free-surface elevation. The initial conditions are specified as:

$$u(x,0) = 0, \quad \theta(x,0) = 4, \quad h(x,0) = \begin{cases} \max(2 - b(x) + \varepsilon, 0), & -1.5 \le x \le -1.4, \\ \max(2 - b(x), 0), & \text{otherwise}. \end{cases}$$

We compute the solution up to $t = 0.4$ with transmissive boundary conditions and compare the results obtained by the scheme with 200 uniform cells, against a reference solution computed using 2000 uniform cells. The computed results displayed in Figure 6 show clearly the scheme's capacity of accurately capturing small perturbations as no oscillations appear over the humps or near wet-dry interfaces. Figure 7 illustrates the temporal evolution of the deviations in water depth and velocity from the reference steady state, namely $h - h_s$ and $u - u_s$, where $(h_s, u_s) = \big(\max(2 - b(x), 0), 0\big)$. The numerical results are reported at times $t = 0.2$ and $t = 0.4$, and are obtained using a uniform mesh of 200 cells. The initial disturbance separates into two wave packets propagating in opposite directions. An additional wave forming near the center arises from the interaction of the initial perturbation with the nearest bottom topography. Throughout the simulation, the solution remains free of spurious oscillations, and the perturbation gradually decays in time. These results emphasize the importance of the well-balanced property of the proposed scheme in the presence of wet-dry interfaces.

4.1.6 Pressure-oscillation-free

In this test case adapted from [4, 14], we assess the well-balanced and positivity-preserving properties of the scheme with respect to the isobaric steady state (1.4) with a temperature jump. The bottom topography is considered to be constant, i.e. $b(x) = 1$, over the computational domain $[-1000, 1000]$. The initial data is defined by

$$(h, u, \theta)(x, 0) = \begin{cases} (2\sqrt{2}, 0, 1), & \text{if } x < 0, \\ (1, 0, 8), & \text{otherwise}. \end{cases}$$

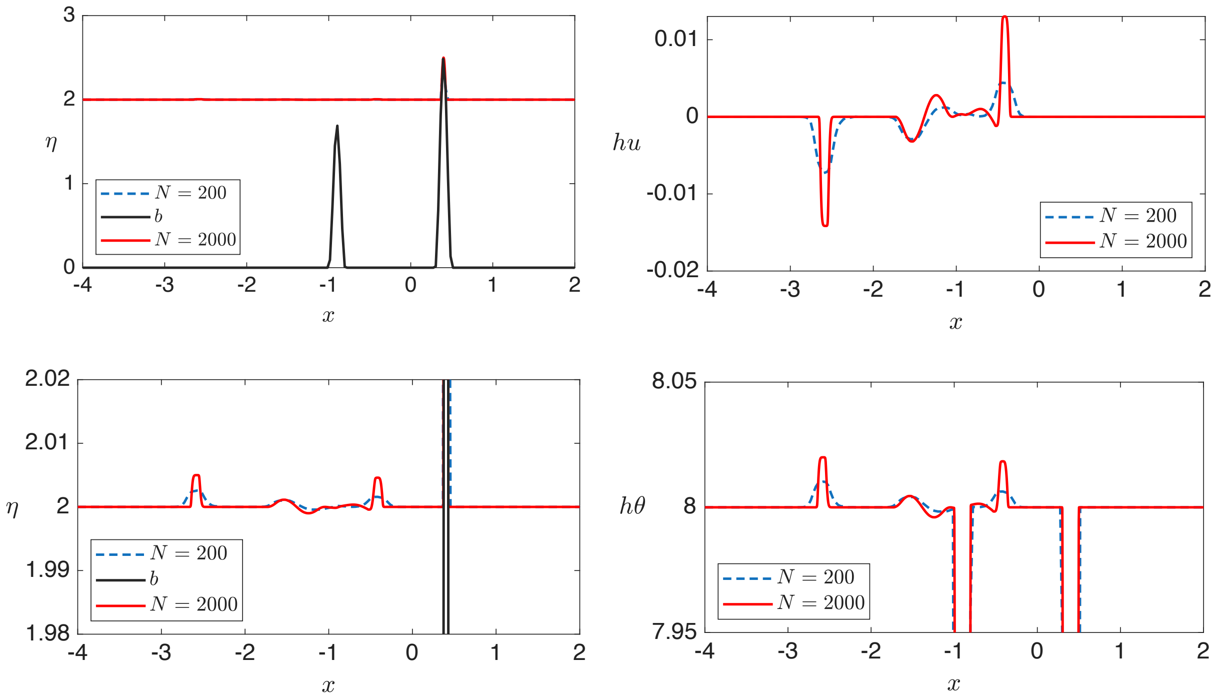


Figure 6: Small perturbations of a steady-state solution. Top left: Water surface elevation. Top right: Discharge. Bottom left: Water surface elevation (Zoom-in). Bottom right: $h\theta$ (Zoom-in).

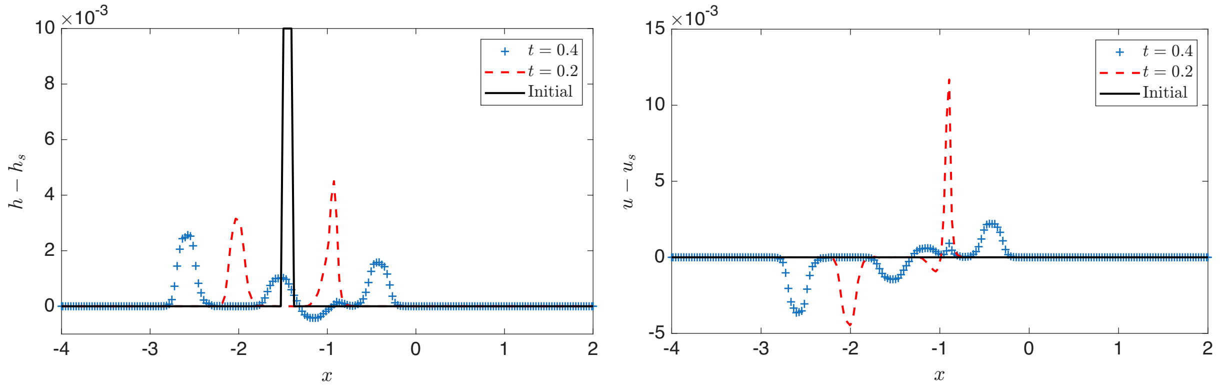


Figure 7: Small perturbations of a steady-state solution. Results in perturbation of the free surface $h - h_s$ (left) and perturbation of the velocity $u - u_s$ (right).

For this configuration, the pressure satisfies $\frac{g}{2}\Pi = \frac{g}{2}h^2\theta = 4$, throughout the computational domain which corresponds to an isobaric steady state (1.4). The presence of a discontinuity in the temperature field defines a configuration known to generate pressure oscillations for non-well-balanced discretizations [4, 14].

In order to highlight the necessity of combining the positivity-preserving parameter of [14] with the proposed discretization of the second source-term component to avoid pressure oscillations in the neighborhood of the temperature discontinuities and preserve the isobaric steady state (1.4), we consider three different numerical schemes in our simulations. We denote by WB-P the fully well-balanced scheme, which combines the positivity-preserving parameter (3.30)-(3.31) with the proper discretization of the second source-term component (3.27). The notation WB-NP refers to the well-balanced scheme in which the positivity-preserving parameter is omitted, while NWB-P designates a non-well-balanced scheme equipped with the positivity-preserving parameter. For completeness, we specify that in the NWB-P scheme, the second source-term component is discretized using the alternative formulation

$$S_{2,i}^{(2)} = -\frac{g}{2}\left(\frac{b_{i+1/2}^n + b_{i-1/2}^n}{2}\right)^2 \left(\frac{\theta_i^{n,+} - \theta_i^{n,-}}{2\Delta x}\right),$$

instead of (3.27). This discretization does not enforce the discrete cancellation required for preserving the isobaric steady state and is used here solely for comparison purposes.

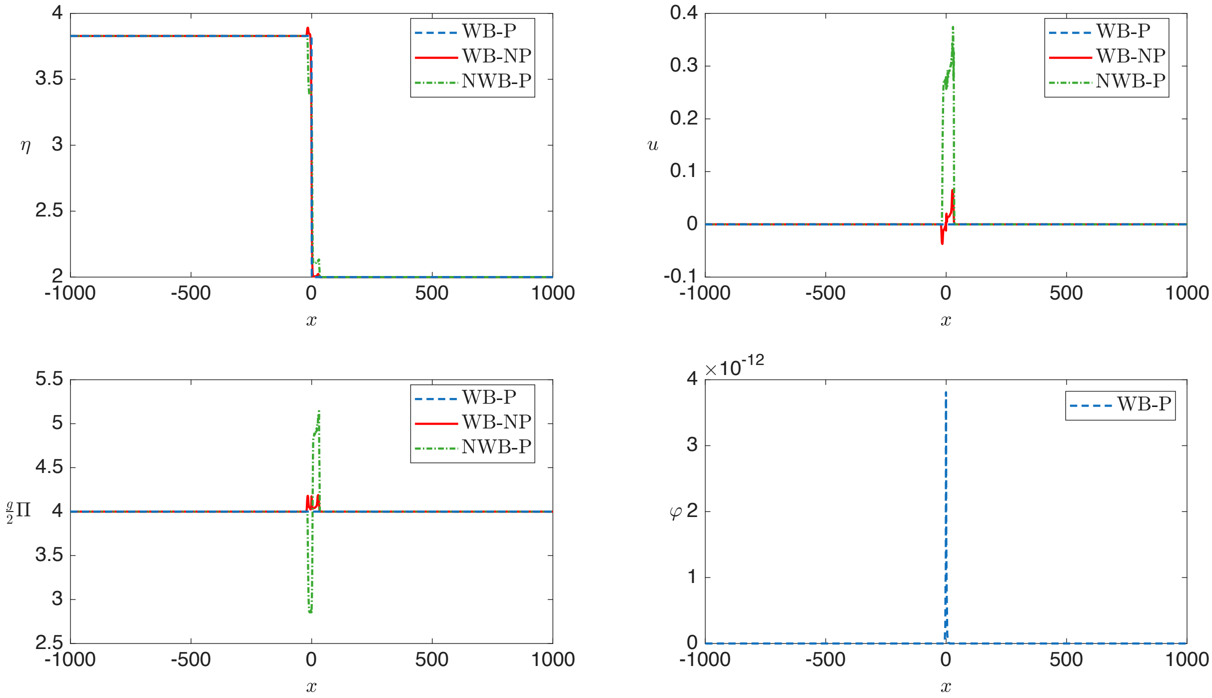


Figure 8: Pressure-oscillation-free. Top left: Water surface elevation. Top right: Velocity. Bottom left: Pressure $\dfrac{g}{2}\Pi$. Bottom right: Computed parameter $\varphi_{i+\frac{1}{2}}$.

The simulation is run until $t = 10$ with 2000 uniform cells and transmissive boundary conditions. The results are shown in Figure 8. As we can see, the computed pressure is oscillation-free when using the WB-P scheme (dashed blue line). To further verify the well-balancing property of the WB-P scheme, we compute the L^∞ norms of the errors in the water surface elevation, the velocity and the pressure with respect to steady-state solution and they turn out to be 7.534×10^{-12}, 2.536×10^{-12} and 6.680×10^{-12}, respectively. In contrast, pressure oscillations are observed when using the WB-NP scheme (solid red line) or the NWB-P scheme (dash-dotted green line). These results demonstrate that neither the positivity-preserving parameter nor the proper discretization of the second source-term component alone is sufficient to prevent pressure oscillations in isobaric steady states.

4.2 Two-dimensional numerical tests

4.2.1 Accuracy test

In this test case, we evaluate the accuracy of the proposed two-dimensional scheme following the setup of [20, 22]. To this end, we consider a smooth bottom topography and the corresponding initial conditions given below:

$$\begin{aligned}
b(x,y) &= \frac{1}{2}\big(\sin^2(\pi x) + \cos^2(\pi y)\big),\\
h(x,y,0) &= \frac{1}{2}\Big(\mathrm{e}^{\cos(2\pi x)} + \mathrm{e}^{\sin(2\pi y)}\Big) - \mathrm{e}^{-1} + 10^{-4},\\
u(x,y,0) &= \sin\!\big(\cos(2\pi x)\big), \quad v(x,y,0) = \cos\!\big(\sin(2\pi y)\big),\\
\theta(x,y,0) &= \frac{1}{2}\Big(\mathrm{e}^{\cos(2\pi x)} + \mathrm{e}^{\sin(2\pi y)}\Big).
\end{aligned}$$

The numerical simulations are carried out on the unit square $[0,1] \times [0,1]$, which is discretized into $N \times N$ uniform Cartesian cells. Periodic boundary conditions are applied in both spatial directions. The initial data are chosen such that the water depth remains strictly positive, with a minimum value of 10^{-4}. The solution is evolved until the final time $t = 0.04$ using the proposed numerical scheme. A reference solution is computed on a fine grid of 800×800 cells. The L^1 errors together with the corresponding experimental orders of convergence are reported in Table 5. These results demonstrate that the scheme achieves second-order accuracy.

Table 5: L^1-errors in h, hu, hv and $h\theta$ along with experimental orders of convergence (EOC) at time $t = 0.04$.

$N \times N$	L^1-error in h	EOC	L^1-error in hu	EOC	L^1-error in hv	EOC	L^1-error in $h\theta$	EOC
25×25	0.0264	-	0.0410	-	0.0205	-	0.0490	-
50×50	0.0064	2.03	0.0101	2.02	0.0055	1.89	0.0121	2.02
100×100	0.0016	1.99	0.0026	1.98	0.0013	2.05	0.0031	1.97
200×200	3.867e-04	2.05	6.140e-04	2.05	3.088e-04	2.09	7.343e-04	2.06
400×400	9.015e-05	2.10	1.453e-04	2.07	7.055e-05	2.13	1.730e-04	2.08

4.2.2 Lake-at-rest steady state with wet-dry interface

This test case, adapted from [20, 22], is designed to illustrate the ability of the proposed scheme to preserve the lake-at-rest steady state in the presence of wet-dry interfaces. The computational domain is $[0,4] \times [0,2]$, and the bottom topography is defined by

$$b(x,y) = \begin{cases} 1 - 0.8\,\mathrm{e}^{-r}, & \text{if } r := 2(x-2)^2 + 4(y-1)^2 < \log\left(\dfrac{8}{5}\right), \\ 0.8\,\mathrm{e}^{-r}, & \text{otherwise,} \end{cases}$$

where r denotes the radial distance from the center of the domain. The initial water depth is prescribed as

$$h(x,y,0) = \begin{cases} \max\big(0.45 - b(x,y), 0\big), & \text{if } r < \log\left(\dfrac{8}{5}\right), \\ \max\big(0.3 - b(x,y), 0\big), & \text{otherwise,} \end{cases}$$

while the initial velocity field is set to zero,

$$u(x,y,0) = v(x,y,0) = 0.$$

The initial temperature is defined by

$$\theta(x,y,0) = 0.2.$$

This configuration partitions the computational domain into two disconnected wet regions separated by a dry zone. Numerical simulations are performed on a uniform grid of 200×100 cells and are conducted up to the final time $t = 0.15$. The numerical results are displayed in Figure 9. The proposed scheme preserves the lake-at-rest steady state up to machine precision, thereby demonstrating its well-balanced and positivity-preserving properties in two space dimensions.

4.2.3 Small perturbations of a steady-state solution

This test case taken from [22] is designed to assess the ability of the proposed scheme to accurately capture small perturbations around a steady-state configuration, with particular emphasis on the interaction between wet-dry interfaces, bottom topography, and temperature variations. We consider a quiescent lake over a nonuniform bottom composed of two Gaussian-shaped humps and defined by

$$b(x,y) = 0.5\,\mathrm{e}^{-100\left((x+0.5)^2+(y+0.5)^2\right)} + 0.6\,\mathrm{e}^{-100\left((x-0.5)^2+(y-0.5)^2\right)}. \tag{4.5}$$

The undisturbed lake-at-rest configuration is characterized by the free-surface elevation

$$\eta^{\mathrm{LaR}}(x,y,0) = \max\big(0.4,\, b(x,y)\big),$$

together with a spatially uniform temperature field given by

$$\theta^{\mathrm{LaR}}(x,y,0) = \frac{4}{3}.$$

To evaluate the response of the scheme to localized disturbances, two distinct types of perturbations are considered: one applied to the free surface and the other to the temperature field. The perturbed initial free-surface elevation (as depicted in Figure 10) is defined as

$$\eta(x,y,0) = \eta^{\mathrm{LaR}}(x,y,0) + \begin{cases} 0.01, & 0.1 \leq \sqrt{x^2+y^2} \leq 0.3, \\ 0, & \text{otherwise}, \end{cases} \tag{4.6}$$

while the perturbed temperature field is prescribed by

$$\theta(x,y,0) = \theta^{\mathrm{LaR}}(x,y,0) + \begin{cases} 0.01, & 0.1 \leq \sqrt{x^2+y^2} \leq 0.3, \\ 0, & \text{otherwise}. \end{cases} \tag{4.7}$$

Numerical simulations are performed on a uniform 200×200 Cartesian grid and evolved until $t = 0.45$. Figures 11 and 12 present the resulting perturbations of the free-surface elevation and temperature, respectively. The results show that the proposed scheme remains free of spurious oscillations near wet–dry interfaces and accurately captures small steady-state perturbations while preserving the positivity of both the water depth and the temperature.

4.2.4 Isobaric steady state

In this test case, we assess the ability of the proposed two-dimensional scheme to preserve the isobaric steady state. We consider a two-dimensional extension of the isobaric equilibrium introduced in [28]. The bottom topography is taken to be constant and the initial data are given by

$$b(x,y) = 1, \quad h(x,y,0) = 1 + \frac{1}{\sqrt{3\pi}} \exp\Big(- \frac{50(x-1.5)^2 + 50(y-1.5)^2}{3} \Big), \quad \theta(x,y,0) = \frac{1}{h(x,y,0)^2},$$

with vanishing initial velocities, $u(x,y,0) = v(x,y,0) = 0$. This configuration satisfies $\Pi = h^2\theta = 1$ throughout the computational domain and therefore corresponds to an isobaric steady state.

The computational domain $\Omega = [0,3] \times [0,3]$ is discretized using a uniform Cartesian grid with 200×200 cells. The simulation is run up to a final time $t = 5$ with periodic boundary conditions imposed in both spatial directions. In Table 6, we report the L^1 and L^∞ errors in the water surface elevation, the velocity components, and the temperature. The results indicate that the proposed scheme preserves the two-dimensional isobaric steady state up to machine precision.

Table 6: Errors for the isobaric steady-state solution.

L^1-error				L^∞-error			
η	u	v	θ	η	u	v	θ
1.765e-13	1.354e-13	1.353e-13	1.124e-13	1.714e-13	1.724e-13	1.644e-13	7.960e-14

5 Conclusion

In this work, we have developed a second-order accurate, well-balanced and positivity-preserving finite-volume scheme for the one- and two-dimensional Ripa system in the presence of wet-dry fronts. The proposed approach is built upon a new pre-balanced reformulation of the governing equations, which embeds the equilibrium structure directly into the flux-source decomposition and enables a consistent treatment of the temperature-dependent pressure term. This reformulation plays a central role in allowing the exact preservation of two physically relevant steady states, namely the lake-at-rest equilibrium, including configurations with dry areas, and the nontrivial isobaric steady state.

At the discrete level, the scheme combines a balanced source-term discretization with a positivity-preserving modification of the HLL flux, yielding stable and oscillation-free solutions across wet-dry interfaces and temperature jumps. High-order accuracy is obtained through MUSCL and hydrostatic reconstructions, while the extension to two dimensions is performed in a dimension-by-dimension framework that preserves the same stability and well-balanced properties under a suitable CFL condition.

A comprehensive set of numerical experiments in one and two space dimensions demonstrates second-order convergence on smooth solutions, preservation of the targeted steady states up to machine precision, and robust behavior in challenging wetting-drying configurations. These results highlight the effectiveness of the proposed scheme and its potential for realistic shallow-water simulations involving coupled temperature effects.

Several directions for future research remain open. These include the construction of schemes preserving additional moving equilibria, the development of higher-order extensions within discontinuous Galerkin or WENO frameworks, and the incorporation of adaptive mesh strategies for large-scale geophysical applications. Another interesting direction for future work is the adaptation of the bottom-surface-gradient reconstruction of [20] to the Ripa system. This would require a coupled correction of the reconstructed bottom and free-surface slopes that remains compatible with the temperature-dependent structure of the model and with the exact preservation of both the lake-at-rest and isobaric steady states.

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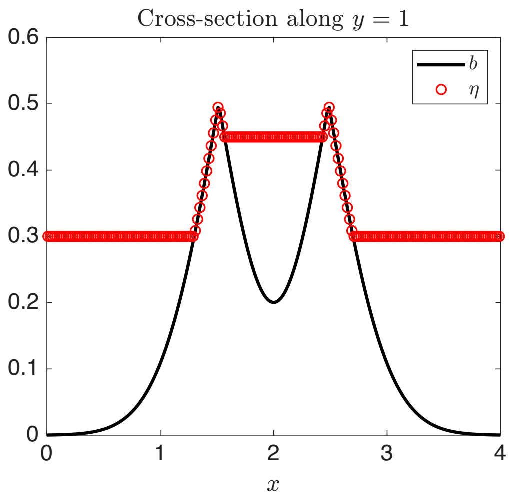
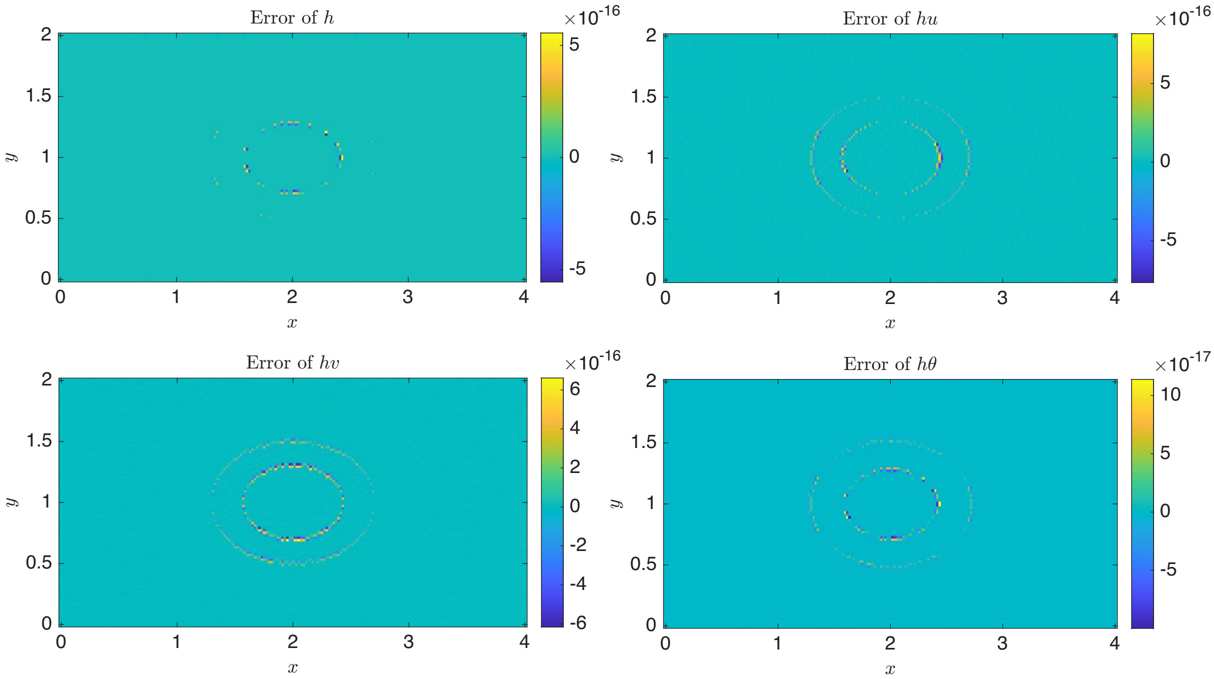


(a) 1-D slice along $y = 1$ of the simulated result.



(b) Numerical errors of h, hu, hv and $h\theta$.

Figure 9: Lake-at-rest steady state with wet-dry interface.

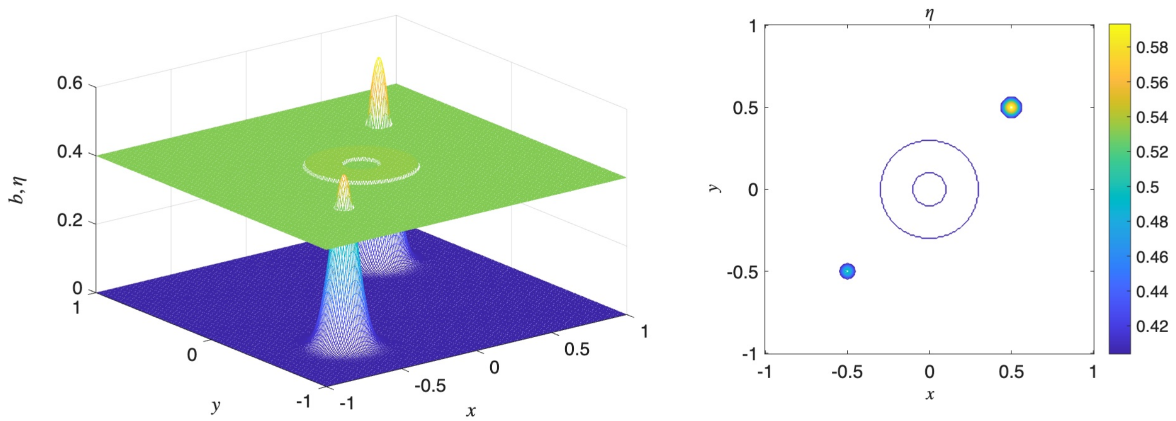


Figure 10: Small perturbation of a steady-state solution. Left: Three-dimensional view of the bottom topography (4.5) together with the perturbed initial free-surface elevation (4.6). Right: Contour plot of the perturbed initial free-surface elevation.

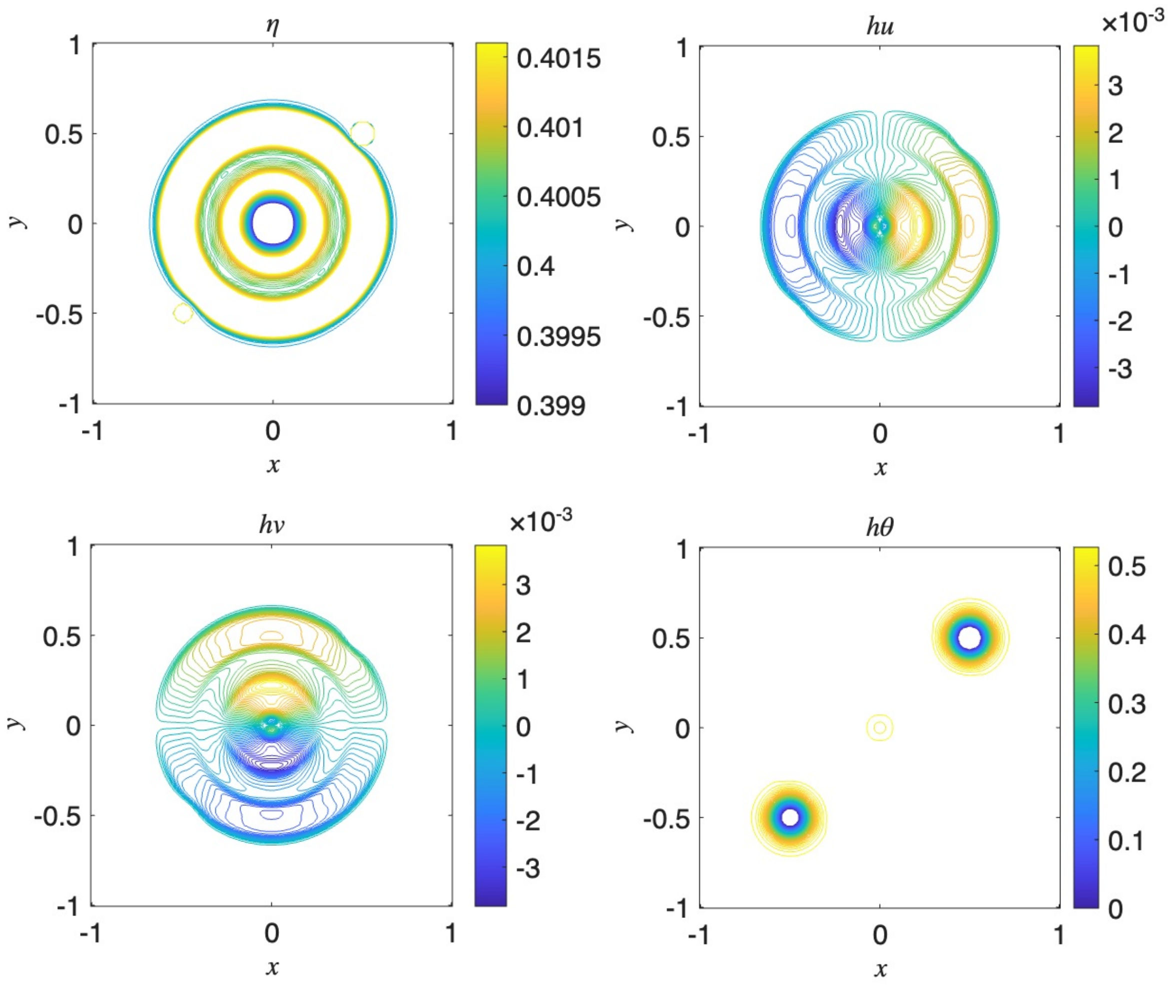


Figure 11: Small perturbation of a steady-state solution. Perturbation is on the water surface elevation (4.6).

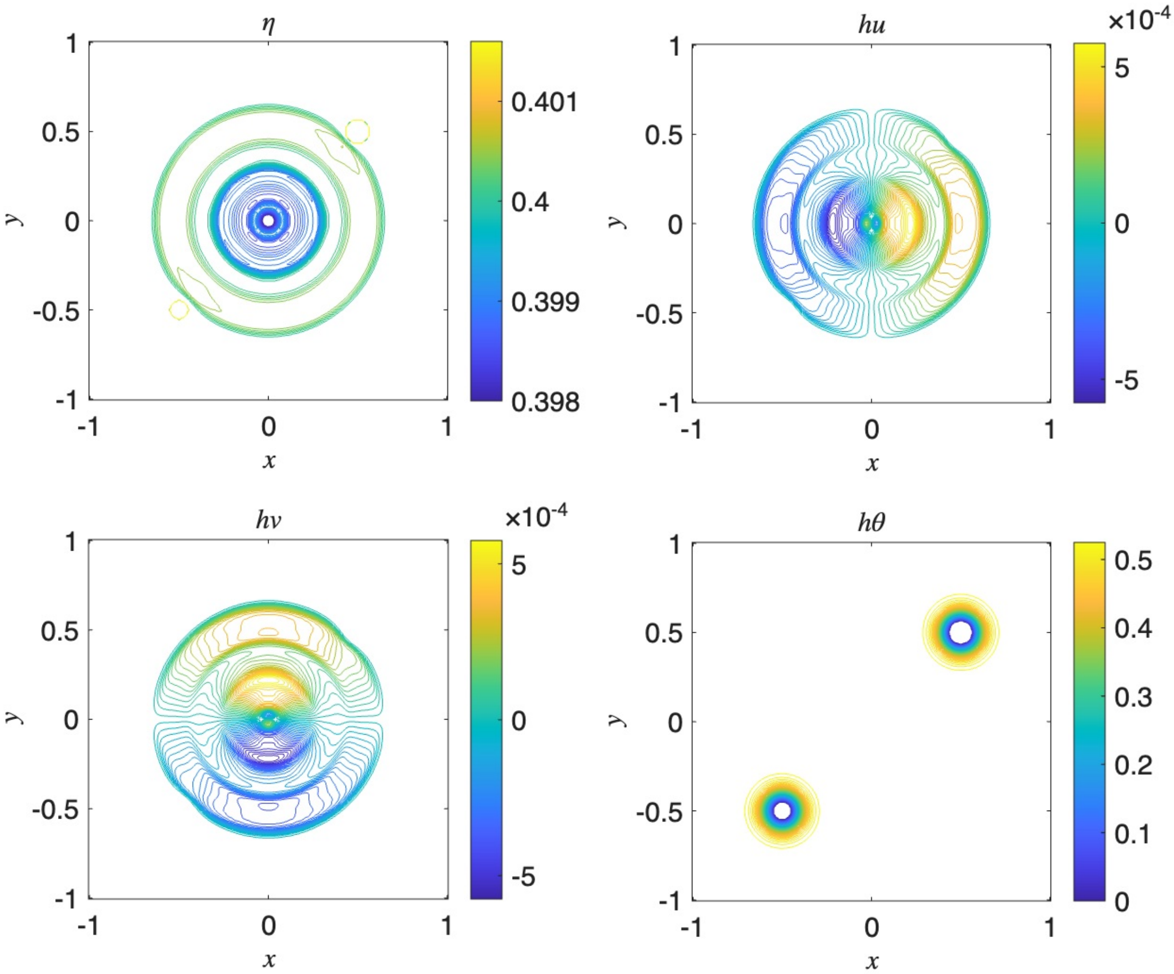


Figure 12: Small perturbation of a steady-state solution. Perturbation is on the temperature (4.7).